ASSIGNMENT 1 FRONT SHEET

Qualification	**BTEC Level 5 HND Diploma in Business**		
Unit number and title	**Unit 30: Application Development**		
Submission date	25/09/2022	**Date Received 1st submission**	
Re-submission Date		**Date Received 2nd submission**	
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Class	GCC0901	**Assessor name**	Pham Truong Hong Ngan
Student declaration I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I understand that making a false declaration is a form of malpractice.			
		Student's signature	Le Hoang Du

Grading grid

P1	P2	P3	M1	M2	D1

❒ **Summative Feedback:**		❒ **Resubmission Feedback:**
Grade:	**Assessor Signature:**	**Date:**
Internal Verifier's Comments:		
Signature & Date:		

Assignment Brief 1 (RQF)

Higher National Certificate/Diploma in Computing

Student Name/ID Number:	**Le Hoang Du/GCC200416**
Unit Number and Title:	**Unit 30: Application Development**
Academic Year:	**2021 – 2022**
Unit Assessor:	**Pham Truong Hong Ngan**
Assignment Title:	**Analysis and Design solution for a given problem**
Issue Date:	
Submission Date:	**25/09/2022**
Internal Verifier Name:	
Date:	

Submission Format:

Format: The submission is in the form of two documents/files

- An individual SRS document in PDF format. Writing must be professional.
- An individual evaluation document in PDF format. Writing must be professional.

Submission

- Students are compulsory to submit the assignment on the due date and in a way requested by the Tutor.
- The form of submission will be a soft copy posted on http://cms.greenwich.edu.vn/.
- Remember to convert the word file into a PDF file before the submission on CMS.

Note:

- The individual Assignment *must* be your work, and not copied by or from another student.
- If you use ideas, quotes, or data (such as diagrams) from books, journals, or other sources, you must reference your sources, using the Harvard style.
- Make sure that you understand and follow the guidelines to avoid plagiarism. Failure to comply with this requirement will result in a failed assignment.

Unit Learning Outcomes:

LO1 Produce a Software Design Document by analyzing a business-related problem and deduce an appropriate solution including a set of initial requirements

LO2 Use design and development methodologies with tools and techniques associated with the creation of a business application

Assignment Brief and Guidance:

Scenario: FPTBook will be a web-based software system to aid book store customers and owners to manage their book purchases and sales about simplifying and speeding up the process of selection, ordering, and purchasing books for customers as well as managing a database of users and a database of products for book store owners. The attached document can view the detailed description of those roles (Requirements.docx).

Tasks

You and your teammates (3 members/team maximum) need to prepare a software design document with the following sections:

- A requirement specification that explores the problem by a set of user and system requirements, as well as determines any risks related to the successful completion of your application. You are advised to use an SRS template or modify one to complete this task.
- An evaluation section in which you research the use of software development tools and techniques and identify any that have been selected for the development of this application. You should compare them and justify your choices.
- A design section in which you use chosen tools from the previous step to produce design diagrams for your solution based on the requirement specification.

Learning Outcomes and Assessment Criteria (Assignment 1):			
Learning Outcome	Pass	Merit	Distinction
LO1	**P1** Explore a business-related problem and produce a well-defined Problem Definition Statement supported by a set of user and system requirements. **P2** Determine any areas of risk related to the successful completion of your application.	**M1** Analyse a business-related problem using appropriate methods and produce a well-structured Software Design Document that defines a proposed solution and includes relevant details on requirements, system analysis, system design, coding, testing and implementation.	**D1** Justify the tools and techniques chosen to realise a custom built website. Justify your preferred selection of tools and techniques in deducing an appropriate solution to a business related problem.
LO2	**P3** Research the use of software development tools and techniques and identify any that have been selected for the development of this application.	**M2** Compare the differences between the various software development tools and techniques researched and justify your preferred selection as well as your preferred software development methodology.	

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1 Design Tools

1.1 Tools to design UML

1.1.1 Lucichart

Making dynamic graphic organizers like seating diagrams, mind maps, Venn diagrams, and more is possible with the help of an online tool called Lucidchart. The capacity of the application to promote cooperation within the cloud-based software allows your complete class (or team) to produce as a unit. Both solitary and collaborative projects benefit greatly from the technology. Real-time collaboration features are provided by the HTML-5-based Lucidchart UML application. It is one of the greatest tools for UML modeling and enables you to build everything from basic technical diagrams to intricate flowcharts (duotl, 2022).

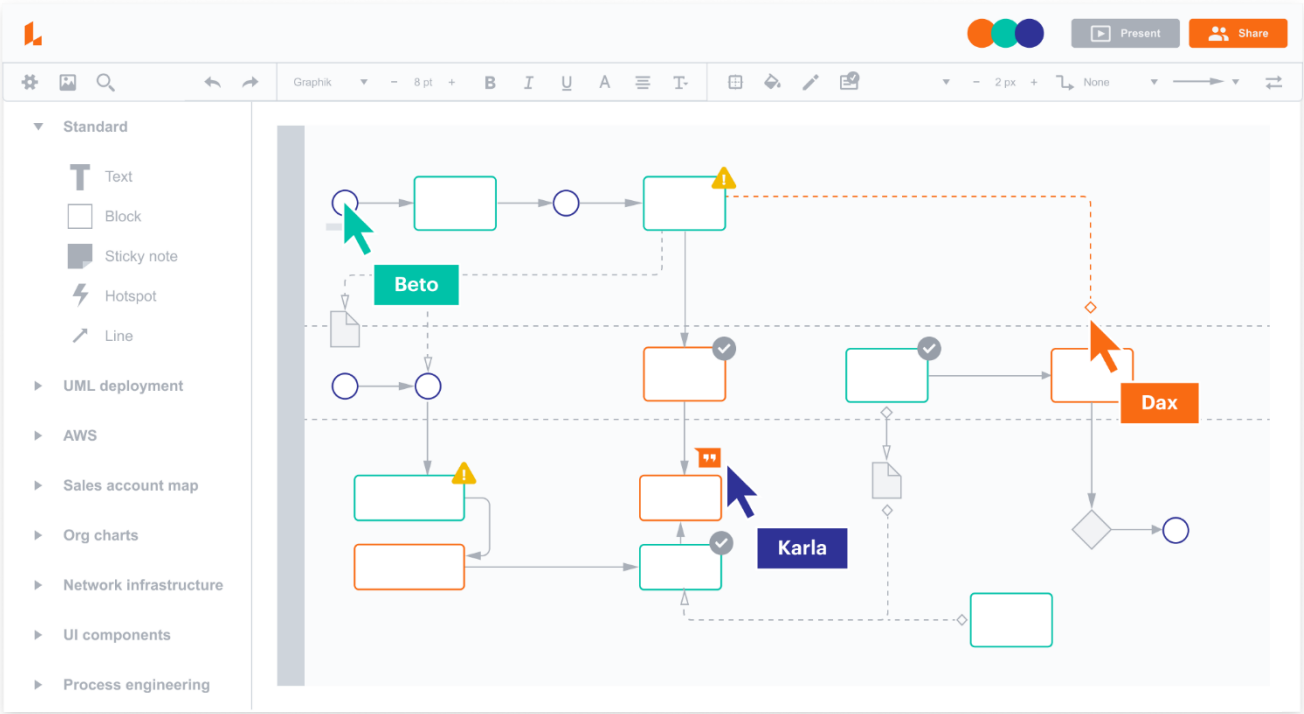

Figure 1: Lucidchart tool[1]

Features:

- Work with your team on many platforms and gadgets.
- Allows you to enter information to automatically generate organizational charts or connect real-time data with your diagrams.
- Increases security and simplifies user account management.
- Microsoft Office, G Suite, Atlassian, and other programs are fully integrated with it.

[1] https://lucid.co/product/lucidchart

1.1.2 Draw.IO

Draw.IO is a free online UML tool. One of the best UML tools, it makes drawing creation and maintenance simple for users. This tool makes a lot of early and widespread sharing feasible. Teamwork is possible with the free, cross-platform desktop version of draw.io for Windows, Mac OS, and Linux. The functions are quite powerful whether it's a flowchart, statistics graph, mind map, or common map. On draw.io, you can draw just about anything you can imagine (Clay, 2021).

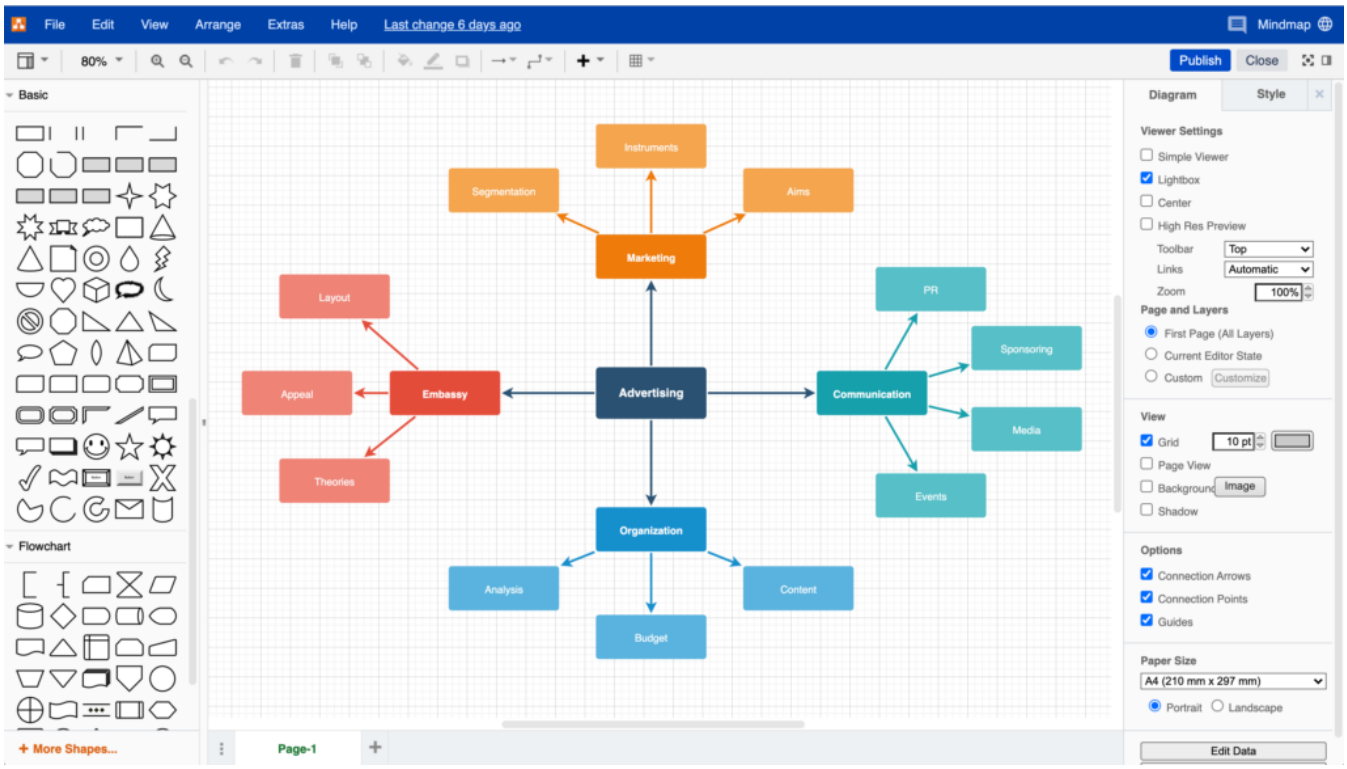

Figure 2: Draw.IO[2]

Features:

- No restriction on the variety of sizes
- Even the design of software uses templates.
- You are able to save the model in your favourite place with our free UML diagram tool.

1.1.3 PlantUML

PlantUML is a free and open source UML diagramming application. It simplifies the development of text-based graphics and is highly efficient to use. The creation of mind maps, flowcharts, class diagrams, object diagrams, and other visualizations is supported. PlantUML was used to easily and instantly preview the flowchart that follows.

[2] https://drawio-app.com/

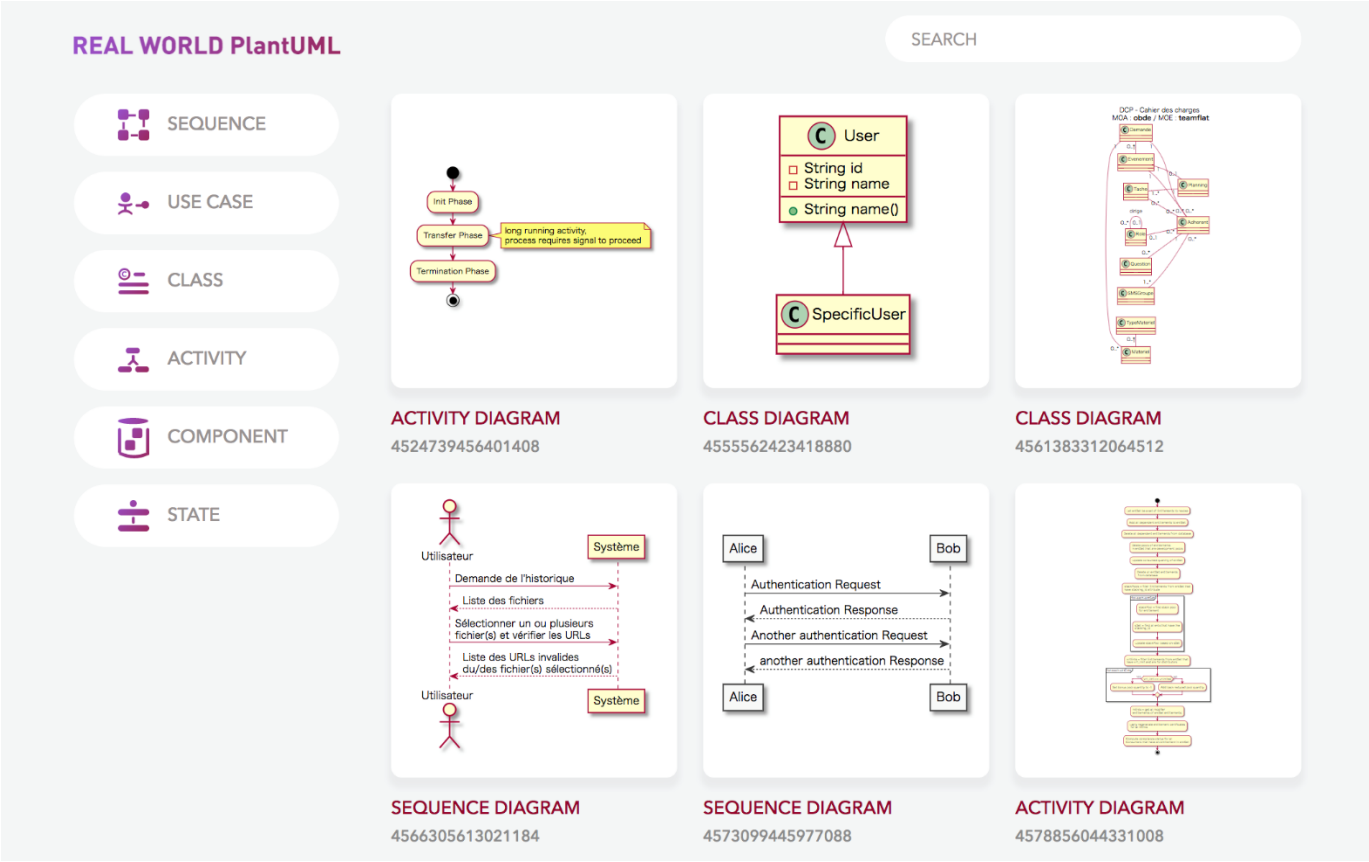

Figure 3: PlantUML[3]

Features:

- The text file size is light, so it's easy to share with others
- Completely free. With most graphic drawing tools that need copyright or free tools, the ability will be limited to some features.
- Since it's text, it's possible to compare changes, review and merge when using git
- No need to install the tool, you can still quickly copy and edit
- Easily export to image files when needed

1.1.4 Comparison between UML design tools

Table 1: Lucichart vs Draw.IO vs PlantUML

UML Tool	Lucidchart	Draw.IO	PlantUML
What is	The well-known US-based diagramming program Lucidchart provides a huge selection of possible shapes and diagram types. It is	Draw.io is a different diagramming tool from the UK that almost anybody can learn in a few minutes. Draw.io is	Users may create diagrams from plain text using an open-source application called PlantUML. PlantUML enables the

[3] https://real-world-plantuml.com/?type=usecase

	easy to use, doesn't need to be downloaded, and integrates with many other programs.	cross-platform and provides a similar variety of templates as Lucidchart. One of the best aspects of Draw.io is that you may get started immediately by visiting their website.	display of JSON and YAML files in addition to a wide range of UML diagrams. It also supports a variety of additional formats relevant to software development, such as Archimate, Block diagram, BPMN, C4, Computer network diagram, ERD, Gantt chart, Mind map, and WBD.
Templates and Shapes	A complete collection of intricate and sector-specific templates, such as wireframes.	On Draw.io, you can create infographics and banners with more creative flare while still having access to all the fundamentals.	The links at the top of plantuml.com demonstrate the variety of diagrams that PlantUML supports. Which type of diagram you wish to generate is specified in the first and last lines of your diagram file to the PlantUML engine.
Importing, exporting, and integrations	File import and export functionality The compatibility of files from several services, like Google, Gliffy, Jira, Confluence, and others, enables third-party integrations.	File import and export functionality The compatibility of files from several services, like Google, Gliffy, Jira, Confluence, and others, enables third-party integrations.	Because PlantUML is a design tool written in code, it can only output pictures, has a low level of file compatibility, and is only useful when the user has access to the source code or needs to code.
User interface	Drag and drop files and shapes onto the diagram using this user-friendly, straightforward interface.	Drag and drop files and shapes into the diagram using this easy, clear interface. You can start using it right away without having to register or download anything.	Can begin instantly without registration or download, has an easy-to-use interface, but users must first get familiar with or master the syntax before using.
Price	The price for Lucidchart is less adjustable yet straightforward. There is a paid version with more features and a larger library, and there is a free version	Draw.io has multiple pricing options, For those wishing to use Draw.io on cloud services, prices range from $15 per month for groups of 20 users, etc. However, if	PlanUML's pricing is simple, it's made for everyone

	with fewer features and no library.	you use it for personal or small-scale purposes, it will be completely free	
Suitable for	Drafts, large projects, teamwork	Suitable for many industries and many users	PlantUML is suitable for drawing diagrams, small projects, and individuals

1.2 Tools to design User Interface

1.2.1 Sketch

Sketch is a vector graphics program that can be used for everything from sketching to wireframing, prototyping, and design handoff. Sketch is a capable and flexible UX and UI design tool built for collaboration. It is widely recognized as an industry-standard tool that is appropriate for both inexperienced and seasoned designers. But keep in mind that Sketch is only compatible with macOS.

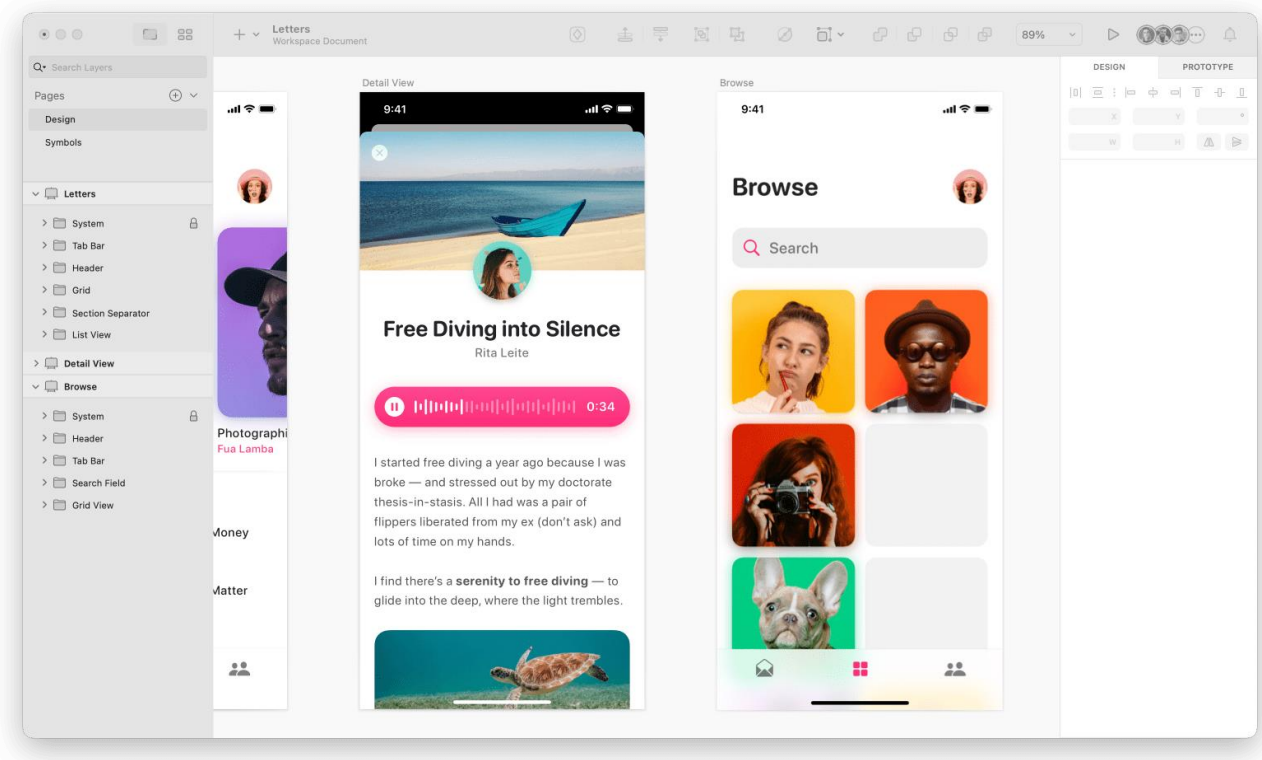

Figure 4: Sketch[4]

[4] https://www.sketch.com/

Features:

- Editable boolean operations and simple vector editing tools enable flexible and iterative design.
- many design possibilities and adaptability Using artboards, design templates, resizable grids, and simple resizing tools, you can scale your designs to any screen size.
- Use mathematical operations and shorthand to speed up the design process.
- Use variable and OpenType fonts for full freedom over your interface typography.
- Platform-agnostic tools for real-time communication, teamwork, and developer handoff.

1.2.2 Adobe XD

Adobe XD is another all-encompassing vector-based UI solution featuring features for group design and prototyping. Adobe XD is frequently recognized as the best design program by users. It is swift, strong, and adaptable, so there isn't much you can't do with it. Adobe XD will provide direction throughout the whole UX and UI design process, from first thoughts and low-fidelity sketches to beautiful animations and accurate prototypes. Because it is a part of the Adobe Creative Cloud suite and works with both Windows and Mac, Adobe XD has a little edge over Sketch.

Figure 5: Adobe XD[5]

Features:

- Intelligent guidelines and an endless artboard drag-and-drop vector editor help you coordinate different objects and elements in your designs.
- Using UI kits (i.e., ready-made components) for Apple Design, Google Material Design, Amazon Alexa, and other 3D Transforms, you may reproduce object depth and perspective in your designs.

[5] https://www.eleken.co/blog-posts/adobe-xd-design-examples-simple-and-inspiring-ideas

- States and elements for an iterative scaling design. By employing components, which work naturally to quickly push changes across whole designs or documents, you may avoid the time-consuming process of manually copying and carrying out changes.
- The Design Specs tool allows you to share interactive prototypes, CSS code snippets, and downloaded documents with developers to hasten the handoff procedure.

1.2.3 Draw.IO

Draw.io is a user-friendly and entertaining diagramming program. Over time, a user will get accustomed to many complex functions that CSS abstracts into visually pleasing intuitiveness. Draw.io competes head-to-head with other well-known solutions like Microsoft Visio or yEd despite the fact that they are each deserving of attention on their own due to their distinct advantages and appealing design. It supports adding custom image libraries and importing schematics from other applications like Visio and Gliffy (Odd Oyvind Andreassen & Patryk Dawid Jankowski, 2020).

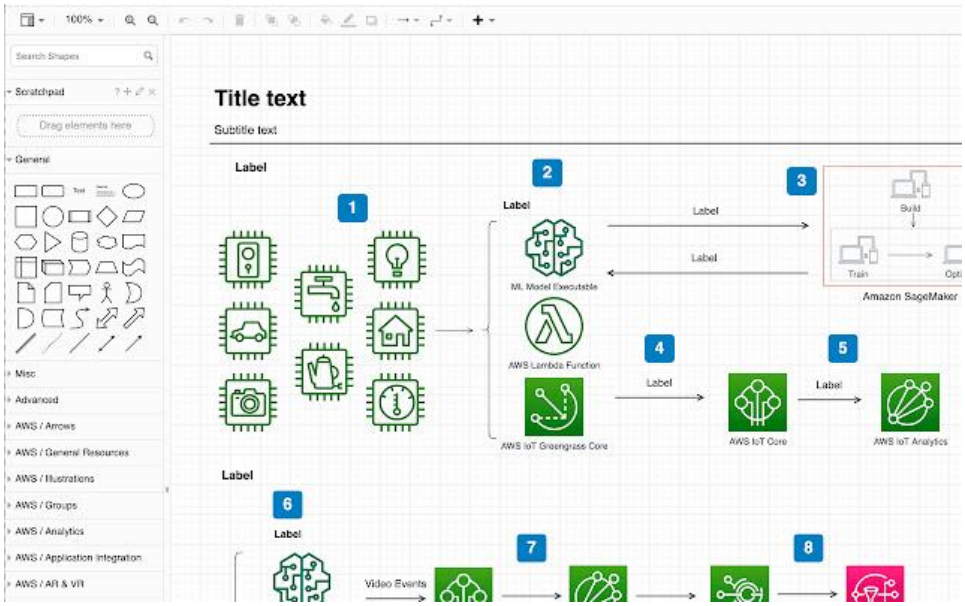

Figure 6: Draw.io

Features:

- Top-rated Confluence diagramming solution since 2013
- Versatile and easy-to-use, with both ready-made and customizable elements
- Works on desktop or mobile instances
- Secure and ISO 9001 compliant

1.2.4 Comparison between User Interface design tools

Table 2: Sketch vs AdobeXD vs Draw.IO

Tool	Skecth	Adobe XD	Draw.IO
Platform	There is software for your computer that you can also use in your browser	Has Desktop and Mobile Apps	There is software for your computer that you can also use in your

			browser. Has Desktop and Mobile Apps
Operating system	Mac OS	MacOS, Windows, iOS, Android	MacOS, Windows, iOS, Android
Collaboration	Real-time collaboration for licensed users using Sketch.	Collaborate in real time on projects synced to the cloud	Collaborate in real time on projects synced to the cloud
User manual	Document	Video tutorials and step-by-step instructions	Document
Additional support library	Large library, need to download to use.	Growing library available in the app	Growing library available in the app
Vector manipulation	Vector path | Vector paths	Vector path | Vector paths	Vector path | Vector paths
Price	Free 30-day trial, then $9/month per user.	Try it for free for 7 days, then $9.99/month.	For those desiring to utilize Draw.io on cloud services, costs start at $15 per month for groups of 20 people, etc. Draw.io includes a variety of pricing choices. However, it will be absolutely free if you use it for private or modest-scale purposes.

1.3 The Design Tools will be used for the development

I decided to use Draw.IO to build our project's UML and user interface due to its advantages and ease. I choose Draw.IO as the design program for our application for the following reasons:

- The desktop version of the free, cross-platform web tool Draw.io facilitates teamwork between users of Windows, Mac OS, and Linux. The functions are highly powerful regardless of whether it's a mind map, common map, statistical graph, or flowchart. On draw.io, almost anything may be produced as long as I can imagine it.
- The range of export formats includes PNG, JPEG, SVG, PDF, XML, HTML, and URL. All of them are artists. I may choose the storage choice from my local device, Google Drive, OneDrive, Dropbox, or GitHub when using the export feature of io.
- I like some of its innovative design in addition to its strong features. When the drawing area is insufficient, for example, another drawing will be automatically added if the object is moved near the drawing's edge; the drawing size does not need to be changed in the settings.
- By visiting the website, I may utilize draw.io right now. By the way, if I want to use draw.io while I am not connected to the internet, I may install the desktop version.

- Draw.io, a powerful, user-friendly, secure diagram-generating tool that is completely integrated with Confluence, makes it simple to draw diagrams.
- Expandable & Flexible, Use pre-made shapes, snapshots, and SVG images. It is simple to drag and drop items directly into the software. Bring in my individual photo archives. Simple import/export of schematics to the precise location where I need them.

Because of the advantages listed below, we select PlantUML tool to generate several diagrams, such as Class diagrams:

- Due to the text file's small size, sharing it with others is straightforward.
- complete freedom The bulk of visual sketching tools that require copyright or are free will have a limited capability.
- Because it's text, I can review, compare, and merge changes when using git.
- Since I may immediately copy and change, there is no need to install the application.
- Export fast to image files when necessary.
- Design speed will be quicker since there is less need to care about component alignment on the drawing diagram. It will take a long time to line up the components' sizes and locations, especially if the diagram has an excessive number of components.
- PlantUML uses plain text and very simple language at that. On the one hand, this expedites writing because I don't have to waste time rearranging lines and boxes or searching through countless options.

2 Front End technology stack

2.1 Front End Programming Language

2.1.1 JavaScript

Open-source JavaScript is a powerful client-side programming language. Its main objective is to improve a user's experience on a website. In other words, you can use this programming language to make your website more vibrant and engaging. JavaScript is frequently employed in the creation of video games and mobile applications. Excellent speed, cross-browser compatibility, and simple semantics are just a few of JavaScript's key attributes that make it an ideal language for developers. The widely used programming language JavaScript enables users to add complex features to websites. It can be activated whenever a page performs a dynamic activity, such as displaying recurring changes to the content, animated two- and three-dimensional graphics, interactive maps, video boxes, and more. JavaScript has first-class functions, prototype-based object orientation, and dynamic typing. It supports event-driven, functional, and imperative programming paradigms, making it a multi-paradigm. It offers application programming interfaces (APIs) for using the Document Object Model, regular expressions, dates, and standard data structures (DOM) (Chinmayee Deshpande, 2022).

Figure 7: JavaScript[6]

Features:

- Object-centered scripting language: Like Visual Basic, JavaScript is an object-centered language with notions based on the idea of objects. When an object may take on numerous forms thanks to a feature called polymorphism, object-centered languages are frequently utilized.
- Client edge technology: Web browsers that are used by users are referred to as clients in client edge technology. A client uploads data to a server, which is subsequently displayed before being made available to users. Users can view webpages and interact with websites using clients or browsers.
- Validation of user's input: Users can start client interactions by filling out of online forms thanks to the validation of user input, often known as form validation. To verify the data entered by a user, the client must check the form's details.

2.1.2 TypeScript

Building on JavaScript, TypeScript is an object-oriented, strongly typed, compiled programming language. In order to provide you with superior tools at any size, it is a superset of the JavaScript language. Anders Hejlsberg, the architect of Microsoft's C#, is the chief architect of TypeScript. TypeScript is a rigorous superset of JavaScript in terms of syntax and gives the language optional static typing. It transpiles to JavaScript and is meant for the creation of huge apps. Existing JavaScript programs are also valid in TypeScript because it is a superset of JavaScript. JavaScript applications may be created with TypeScript for server-side and client-side execution (as with Node.js or Deno). There are several transpilation alternatives. The default TypeScript Compiler can be used, or the Babel compiler can be invoked to convert TypeScript to JavaScript (Meredith Shubel, 2022).

[6] https://www.geekboots.com/story/importance-of-javascript-in-web-development

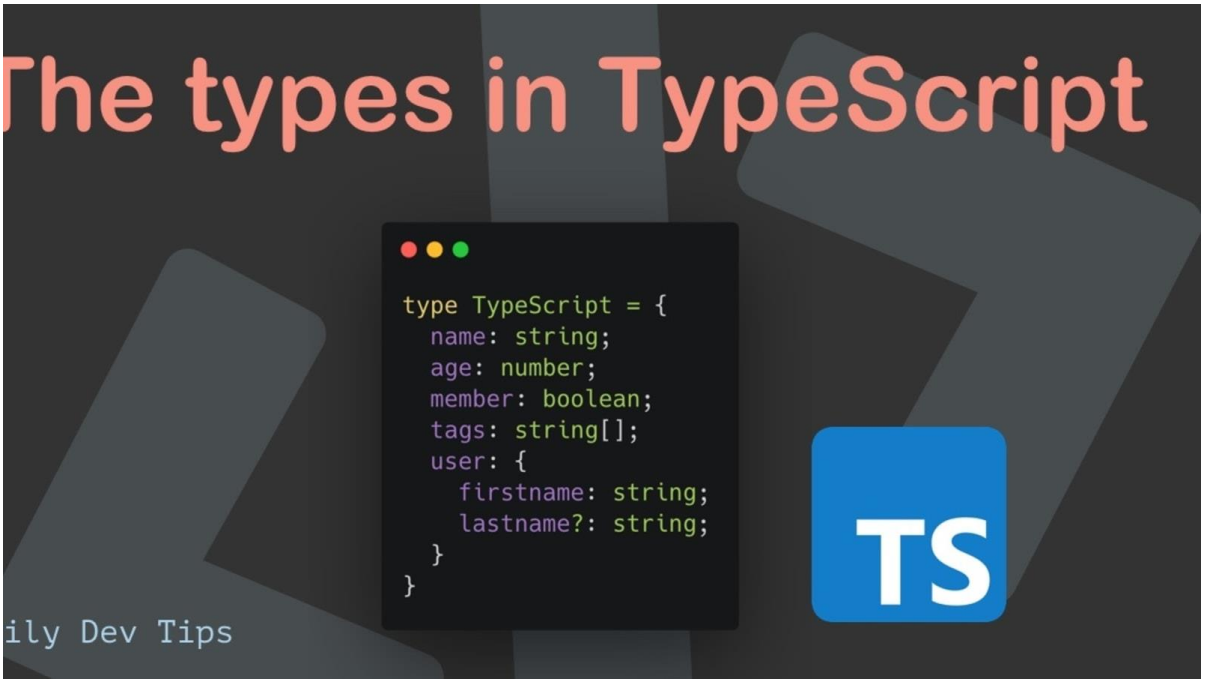

Figure 8: TypeScript[7]

Feature:

- TypeScript supports all JavaScript libraries and frameworks. Angular, a JavaScript framework used to build single-page applications is entirely written in TypeScript.
- With an increase in code complexity, JavaScript had to fulfill the requirements of OOP; hence that lead to the introduction of TypeScript.
- TypeScript helps with quicker code development, thus improving performance.

2.1.3 Hypertext Markup Language (HTML)

The computer language that makes it easier to create websites is called HyperText Markup Language. The language is comparatively simple to understand and grows more powerful in what it allows someone to produce with time. It features code terms and syntax much like other languages. Under the direction of the World Wide Web Consortium, the company that created and maintains the language, HTML keeps changing to suit the needs and specifications of the Internet, such as with the shift to Web 2.0. Users of the Internet navigate the web via hypertext. Users are directed to new sites by clicking on hyperlinks—special text that can be found in the text. Users can access any page on the Internet by clicking on one of the accessible links because the usage of hyper implies that it is not linear. HTML tags designate the content they contain as a certain sort of text by applying markup to it. To specifically highlight a word or phrase, markup language can take the shape of boldface or italicized font, for instance (Adam Hayes, 2020).

[7] https://thenewstack.io/what-is-typescript/

Figure 9: HyperText Markup Language[8]

Features:

- HTML is a programming language with an easy learning curve and offers easy usability.
- The language is also platform-independent.
- HTML facilitates the easy addition of videos, images, and audio content to a web page.
- The language can also be used for adding hypertext content to sections of text.
- HTML is a markup language and a lot less complex than other languages.

2.1.4 Cascading Style Sheet (CSS)

Despite being a style sheet, CSS is regarded as the best front-end language. Cascading Style Sheets, or CSS, is a design language that makes it easier to present web pages. Selectors are tools for choosing items and styling them. A document may be formatted in a way that audiences can comprehend using CSS. This is crucial for browsers like Chrome, Firefox, and Edge which are made to present content graphically to screens, printers, and projectors. CSS is frequently used because it gives selectors, text effects, and layouts, saves time during development and is simple to maintain. Layout, color, and font may all be separated from text and presentation using CSS. By specifying the pertinent CSS in a separate.css file, which reduces complexity and repetition in the structural content, this separation can improve content accessibility, provide more flexibility and control in the specification of presentation characteristics, enable multiple web pages to share formatting, and enable the.css file to be cached to improve page load speed between the pages that share the file and its formatting (Chinmayee Deshpande, 2022).

[8] https://vietnambiz.vn/ngon-ngu-danh-dau-sieu-van-ban-hypertext-markup-language-la-gi-dac-diem-20200412155545061.htm

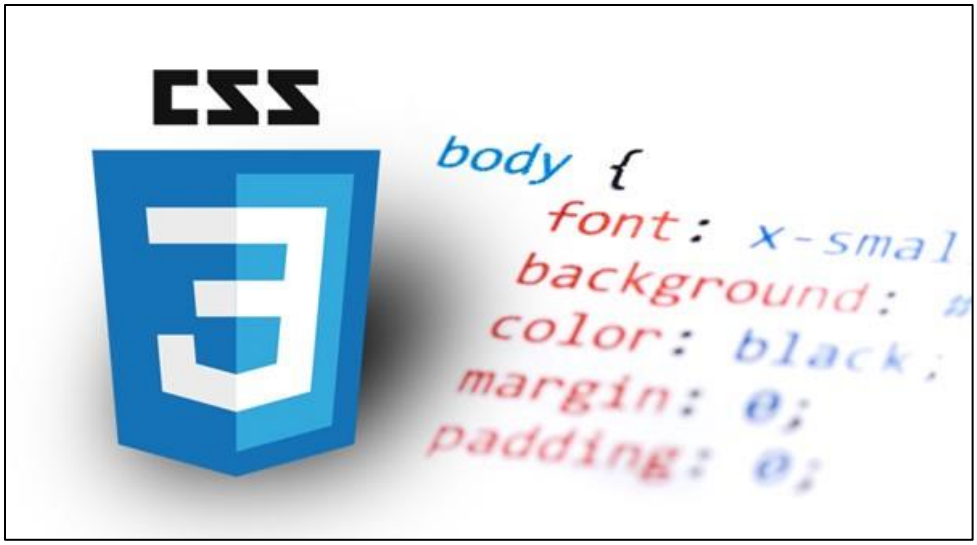

Figure 10: CSS[9]

Features:

- Selectors: Users may choose and manage various components on a web page using CSS selectors. They may be thought of as structural elements for carrying out attribute and attribute value matching. Pseudo-classes may now be targeted by new selectors to style items that are URL-targeted. A checked pseudo-class is also included in selectors for styling checked components, such as checkboxes and radio buttons.
- Text effects and layout: The whitespace in a document may be changed, text justification can be changed, and word hyphenation styles can all be changed using CSS3.
- First-letter and first-line pseudo-classes: CSS has features that make it easier to modify character spacing to improve the aesthetic impact and add drop caps to the start of paragraphs.

2.1.5 The comparison of the front-end programing languages

JavaScript vs TypeScript:

Table 3: JavaScript vs TypeScript

Parameter	Typescript	JavaScript
What is	Powerful type system with features like generics and JS	Lightweight, interpreted, object-oriented language with first-class functions
Data Binding	TypeScript describes the data being utilized by using ideas like types and interfaces.	No such concept is available with JavaScript.
Ecosystem	The Ecosystem is really strong and logical. This makes it possible to statically type a variety of idiomatic JavaScript features, such as union types, intersections, and discriminated unions.	JavaScript offers the option to explore and create code without a build step.

[9] https://webdesignservice.sg/css-an-introduction-to-css/

Npm package	Many npm packages for Typescript either provide an external type definition that is simple to install or come with static type definitions.	JavaScript offers the option to explore and create code without a build step.
Learning curve	challenging learning curve. calls for prior scripting experience.	Flexible and easy to learn, scripting language.
Prototyping	A characteristic of Typescript is prototyping.	JavaScript doesn't have this feature.
Community	The developer community for Typescript is small.	The JavaScript has a huge community of developers

HTML vs CSS:

Table 4: HTML vs CSS

HTML	CSS
HTML is a markup language used to define a structure of a web page.	CSS is a style sheet language used to style web pages by using different styling features.
It consists of tags inside which text is enclosed.	It consists of selectors and declaration blocks.
HTML doesn't have further types.	CSS can be internal or external depending upon the requirement.
We cannot use HTML inside a CSS sheet.	We can use CSS inside an HTML document.
HTML is not used for presentation and visualization.	CSS is used for presentation and visualization.
HTML has comparatively less backup and support.	CSS has comparatively higher backup and support.
HTML doesn't allow animations and transitions.	CSS allows animation and transitions which helps to improve the UI.

2.2 JavaScript Library / Framework

2.2.1 Angular

Single-page web apps are typically created using the open-source framework Angular. It may be relied upon by developers to produce animated menus suited for HTML websites. It is a framework created by Google developers Misko Hebert and Adam Abrons. AngularJS, which was initially launched in 2012, is continually updated. Angular is a TypeScript-based open-source JavaScript framework. It is managed by Google, and one of its key objectives is to create single-page apps. Angular is a framework that clearly has benefits, and it also gives a team of developers a consistent framework to work with. It enables users to easily manage large-scale applications. Angular is an opinionated framework which means that it specifies a certain style and certain rules that developers need to follow and adhere to while developing apps with Angular, therefore you need to learn Angular and the various components that make up an Angular app (Chinmayee Deshpande, 2022).

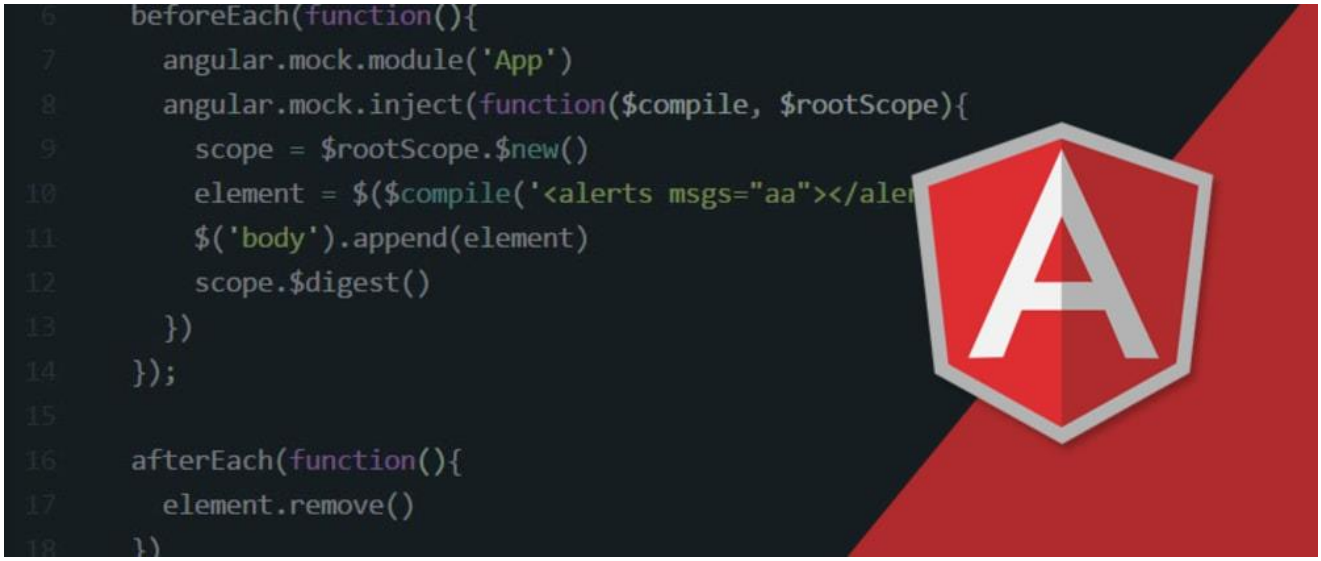

Figure 11: Angular[10]

Features:

- Data Binding: It employs dynamic HTML and does not require complex scripting or programming.
- TypeScript: Angular makes use of TypeScript. Users may build more intelligible JavaScript code by using a collection of types called TypeScript. JavaScript may be used to compile all TypeScript code, allowing it to execute on any platform.
- Progressive web apps: Developers can use cutting-edge platform features to create similar experiences to apps. Angular offers high performance and easy installation.
- Angular CLI: Angular offers command-line tools to enable developers to create apps faster, add tests, and achieve fast deployment.

2.2.2 React

React is a JavaScript-based UI development library. React is widely used in web development even though it is a library and not a language. The React front-end library, which was released in May 2013, is presently one of the most used ones for web development. For the purpose of developing quick and innovative web user interfaces, Facebook has created React. Currently, it is a top JavaScript library for building online front ends and a fantastic choice for full-stack development. React is an appropriate programming language for building graphical user interfaces and displaying data for browsers. With React, front-end web development is simple, and the following sections go through the main benefits and features of this technology. React goes beyond simple UI and has many extensions like Flux and React Native for complete application architecture support (Chinmayee Deshpande, 2022).

[10] https://dev.to/manas_dev/angularjs-is-dead-455n

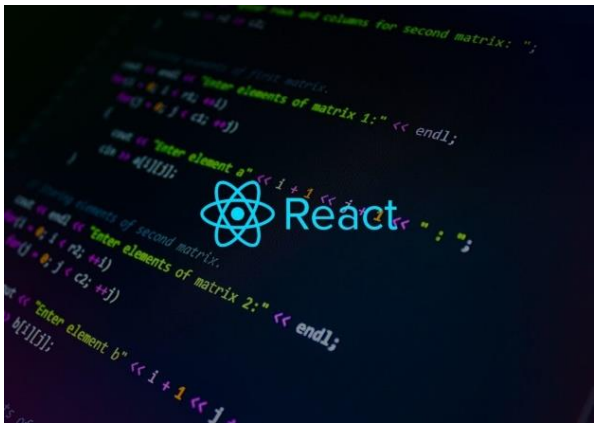

Figure 12: React[11]

Features:

- JSX: JSX is among the syntax extensions of JavaScript. It is utilized with React Native for describing the appearance of a user interface. Using this extension lets users create HTML structures that exist in the same file as JavaScript code.
- Virtual DOM: React stores a lighter version of DOM within the memory, which can be referred to as VDOM or virtual DOM. Users are able to tweak VDOM faster than DOM due to there not being a need for screen drawing.

2.2.3 VueJS

Vue, a JavaScript framework for building user interfaces, differs from other monolithic frameworks in that it was developed to provide the most possible flexibility. The primary Vue library is concentrated on a view layer and is simple to use in addition to being simple to connect with other project libraries already in place. Declarative rendering and component composition are the main foci of Vue.js' progressively flexible design. The view layer is the only area of attention for the core library. Routing, state management, and build tools are examples of advanced functionalities made available via officially maintained supporting libraries and packages. By adding directives, which are HTML attributes, Vue.js enables the extension of HTML. The directives, which can be either built-in or user-defined, provide the functionality to HTML applications (Chinmayee Deshpande, 2022).

Figure 13: VueJS

[11] https://beeproger.com/blog/react-wat-is-het-en-hoe-werkt-het/

Features:

- Virtual DOM: Virtual DOM, which is effectively a replica of the primary DOM component that responds to each change in the DOM, is a feature of the Vue.js framework and is available as a JavaScript data structure. The evolution and modifications of JavaScript data structures may be compared to their initial forms. The last updates that consumers can see are applied to the actual DOM.
- Two-Way data binding: Due to its MVVM design, VueJS has two-way binding. Utilizing the QRR facilitates editing or allocating HTML value attributes to speed up HTML blocks. To make sure that all UI changes are transmitted to the UI, VueJS includes two-way communication.

2.2.4 The comparison of the JavaScript Library / Framework

Table 5: React vs Angular vs VueJS

Categories	React	Angular	VueJS
Type	Rich Library to build UI	Framework	Framework
Development Approach	Everything is on JavaSript	Based on TypeScript	Based on JavaScript and HTML
Maintained and Supported By	Facebook	Google	Former Google Employee
Initial Release	2013	2016	2014
When to Choose	React provides you with modern widgets and in - built features that help you develop SPA and mobile apps for different platforms using a single codebase.	Undoubtedly, Angular is a mature framework, so if you want to develop native apps, hybrid apps , and web apps, Angular is good to go.	Offering a wide choice of widgets that enables you to build advanced SPA and start supporting Native apps.
Ideal for	Suitable for modern web development and native - rendered apps for iOS and Android	Angular is an ideal approach to building large - scale, feature-rich applications or enterprise - level apps.	Ideal for web development and single - page applications
Learning Curve	Easy to access as it is based on JavaScript	A steep learning curve	A small learning curve
Developer-friendly	Ensuring a flexible development environment	It embraces structure-based framework	In case, your focus is on separation of concern
Model	Based on Virtual DOM (Document Object Model)	Based on MVC (Model-View-Controller) architecture	Based on Virtual DOM (Document Object Model)

Written in	JavaScript	TypeScript	JavaScript
Language Preference	Recommends the use of JSX – JavaScript XML	Recommends the use of TypeScript	HTML templates and JavaScript

2.3 CSS Framework

2.3.1 Bootstrap

Bootstrap was developed by Jacob Thornton and Mark Otto at Twitter as an open-source framework containing CSS and JavaScript-based templates for interface components in order to promote uniformity among internal tools. Bootstrap is to blame for web developers' concentration on responsive design. It promoted the now-common notion of "mobile-first" and provided the tools needed for its straightforward implementation. In order to do this, it divided the screen into invisible (to the user) columns and added a grid. With Bootstrap, developers can scale down a website without starting new projects just for that purpose. If the proper Bootstrap classes are provided, the design will automatically adjust (Shreya Bose, 2022).

Figure 14: Bootstrap[12]

Features:

- Massive Ecosystem: The front-end framework Bootstrap has an unmatched ecosystem. It gives developers and designers access to a vast library of pre-built layouts, themes, UI components, panels, modals, buttons, alerts, cards, etc. Furthermore, Bootstrap offers the best community support in the industry.
- Accelerated prototyping: Web designers may create responsive websites with Bootstrap by simply typing their HTML code and adding the required CSS classes. Adjustments for things like CSS positioning and browser compatibility don't require them to invest much work.

[12] https://wiki.matbao.net/bootstrap-la-gi-cai-dat-bootstrap-web-chuan-responsive/

- Twitter Support: Unsurprisingly, when a major commercial entity supports an open-source initiative, customers may feel secure that it is here to stay and enjoys high confidence among individuals in the business. The fact that Bootstrap was inspired by and is backed by Twitter serves as evidence of its success.
- Supports SASS and LESS: The majority of developers do not utilize LESS, although significant projects do. It goes without saying that SASS support is also highly desired. Few CSS frameworks, aside from Bootstrap, include both.

2.3.2 Tailwind CSS

Tailwind It is described as a "utility-first CSS framework" in the official CSS documentation because it has classes that let users directly create distinctive user interface designs in their markup. It is simple to utilize inline style to build a lovely user experience without having to develop any CSS. Tailwind One of the most popular utility CSS frameworks, CSS, provides a number of benefits for web design. Oddly enough, it took Tailwind's Adam Wathan to convince everyone that utility-based CSS is better than semantic CSS. However, as soon as enough programmers believed him and started using Tailwind, it became popular enough to be listed on lists like these (Shreya Bose, 2022).

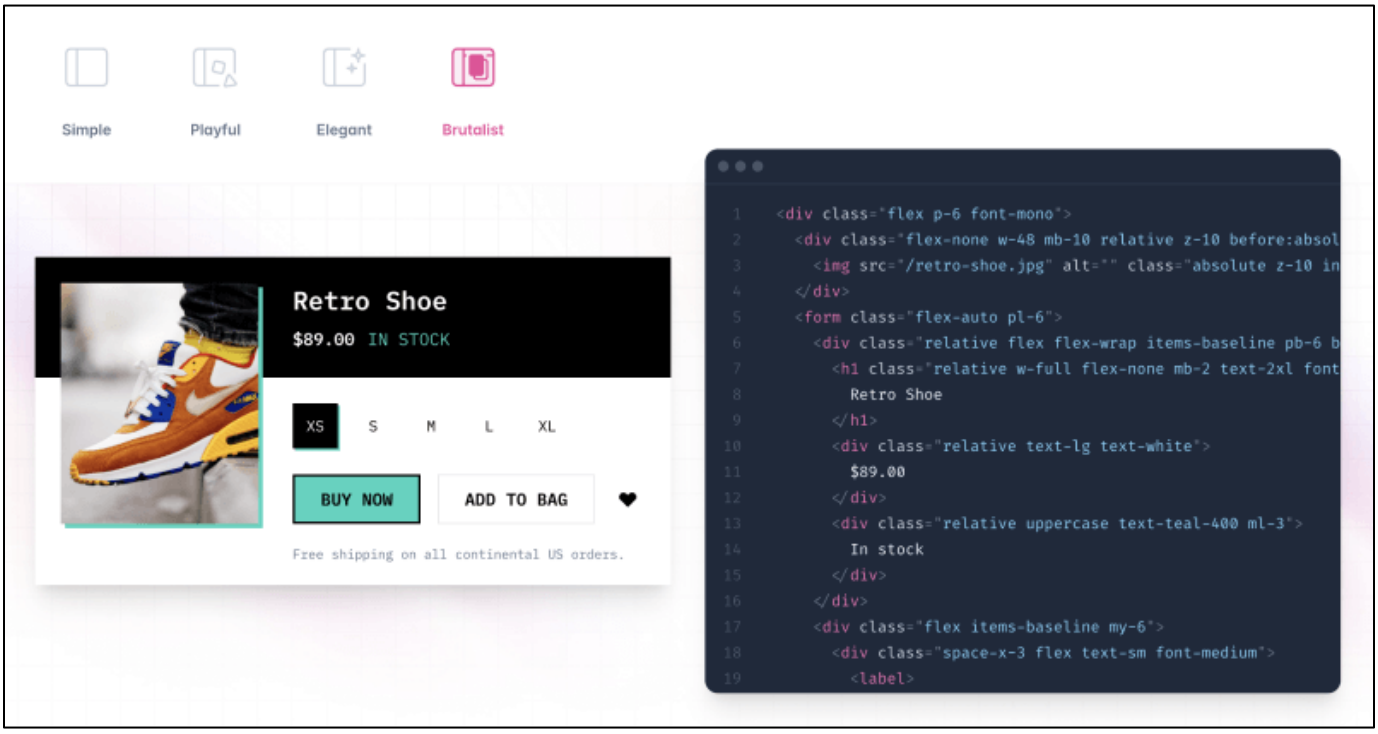

Figure 15: Tailwind CSS[13]

Features:

- Highly Customizable: The Tailwind CSS default configuration may be overridden with a tailwind.config.js file. Because of this, changing the style, themes, spacing, palettes, etc. is straightforward. Create a user-friendly website and simplify project management by using Tailwind's features.

[13] https://unitop.com.vn/tailwind-css.html/

- Carries Common Utility Pattern: Users can avoid naming too many classes by utilizing Tailwind CSS. To address common requirements, specifying and organizing classes, cascading classes, and other widespread utility patterns are given. Simply told, it becomes much easier to design new components. Instead of hard-coding variables, just use the theme() method to retrieve them from configuration files.
- PurgeCSS Optimization: By using Tailwind CSS, users may avoid naming too many classes. Cascading classes, describing and arranging classes, and other common utility patterns are provided to fulfill common requirements. Simply said, designing new components becomes considerably simpler. Use the theme() function to get variables from configuration files instead of hard coding them.
- Responsiveness made easy: Tailwind CSS defaults to a mobile-first approach. According to the website, any utility class in Tailwind can be applied conditionally at different breakpoints, making it simple to create intricate responsive interfaces without ever leaving your HTML. Utility classes facilitate the development of complex responsive layouts while maintaining their usability at different breakpoints. The end result is the easy deployment of responsive design.

2.3.3 Foundation

The "most cutting-edge responsive front-end framework in the world" is how Foundation describes itself. This responsive front-end framework offers a grid, templates, code, and UI components for navigation, buttons, typography, forms, etc. in HTML, SASS, and CSS. It also includes extra features provided via JavaScript extensions. As of 2019, the open-source project Foundation is run by volunteers rather than being managed by ZURB. It also employs a mobile-first approach and is particularly beneficial for building large web apps that require a design host (Shreya Bose, 2022).

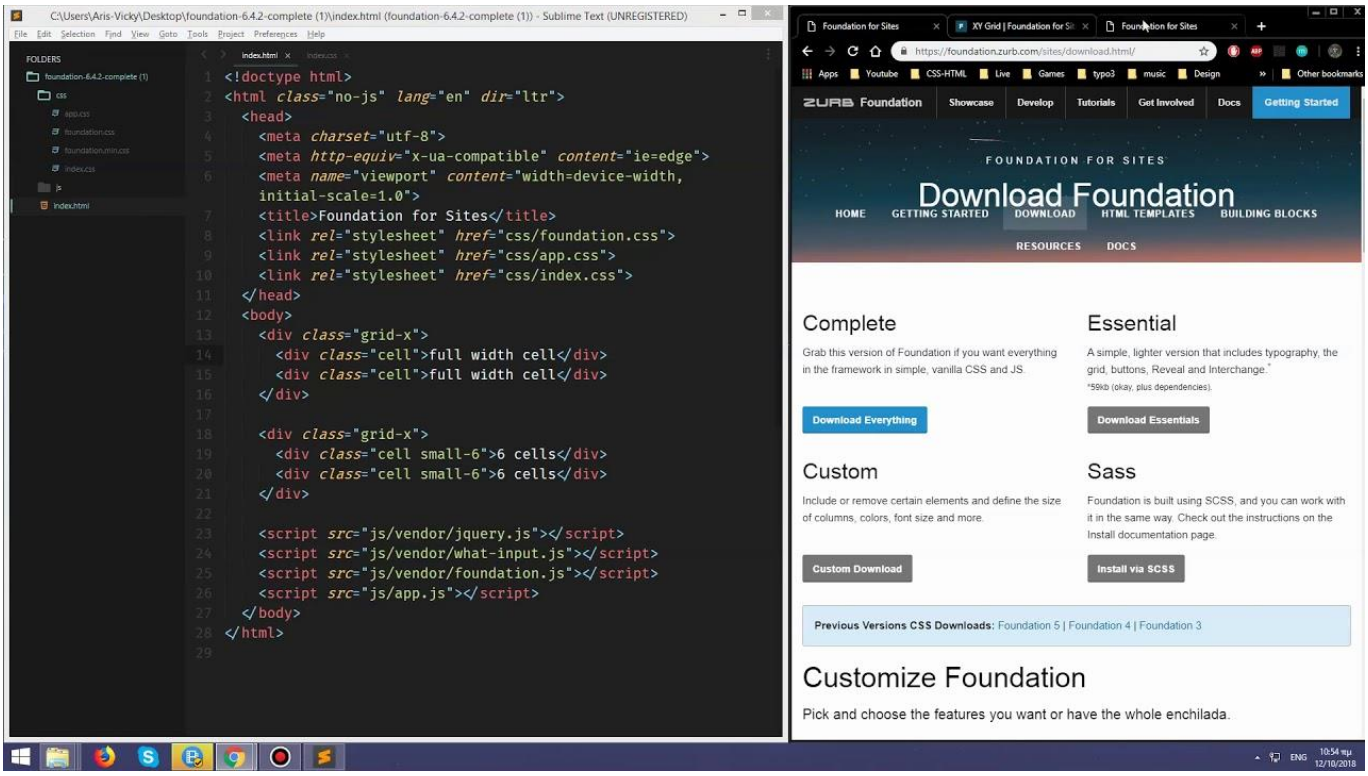

Figure 16: Foundation[14]

[14] http://www.gklondon.co.uk/foundation-js-k.html

Features:

- Massive toolkit: Foundation offers more than what is required in a CSS framework. The majority of front-end developers discover answers to issues because of the extensive, modular toolset it provides. For instance, it offers separate, ready-to-use core components for emails and web pages, respectively. Additionally, developers that use module bundlers like Webpack can benefit from the availability of a command-line interface (CLI).
- Greater Flexibility: Foundation was developed to provide front-end developers complete control over their UIs. For instance, it doesn't mandate that developers use a certain language or literary style. Because of all of this, a lot more developers prefer it as a tool.
- UI components and beyond: In addition to the typical UI elements, Foundation offers a responsive image system, a pricing table, responsive embeds, form validation, right-to-left compatibility, and much more. This gives website designers a far wider range of resources.
- Training Options: For Foundation, ZURB provides a variety of training programs and consultancy services, which is especially useful for businesses looking to invest in large-scale projects.

2.3.4 The comparison of the CSS Framework

Table 6: Bootstrap vs Tailwind CSS vs Foundation

Tool	CSS Tailwind	Bootstrap	Foundation
What is	The user experience of a website may be instantly customized using Tailwind's predesigned widgets.	With Bootstrap, you get a selection of mobile-first, responsive components with a particular UI kit.	A responsive grid, HTML and CSS UI components, templates, code snippets, and optional JavaScript extensions are all included in the free and open-source responsive front-end framework Foundation. Typefaces, forms, buttons, navigation, and other interface elements are some of these elements.
Convenience	Tailwind CSS makes use of a variety of utility classes to create a simple UI with greater adaptability and personality.	Bootstrap-built websites follow a consistent design pattern, giving them a unified look.	To create a flexible environment for design, Foundation combines various control tools with the web design languages CSS and basic HTML.

Customization	Website modification is far more flexible using Tailwind CSS.	Despite the fact that websites created using Bootstrap are recognized for their responsiveness and impeccable design, the aesthetics are uniform and comparable.	It supports a very special and simple phone menu interface and has numerous practical Javascript effects. Additionally, the Foundation Framework's UI style is created in a widely used flat form, making it simple to utilize for flat interfaces.
Effective	Even while Tailwind CSS is still relatively new and has a lot of potential for community growth, it continues to grow daily in terms of users, websites, tools, and references.	Utilizing Bootstrap is efficient and saves time. Since it has been in use for more than nine years and is the most well-known CSS Framework, it has a substantial development community as well as forums, tools, and other resources.	Instead of only utilizing CSS as a distinct online design language, developers came up with the notion of combining Foundation and other Sass/SCSS tools to optimize all web design tasks fast and effectively.
Capacity	Because it just needs the base stylesheet file, which is only 27 kilobytes in size, Tailwind CSS is easier to utilize.	You need to incorporate all four of Bootstrap's files in your project in order to properly utilize the CSS Framework. When you include jQuery, Popper.js, Bootstrap JS, and the main Bootstrap CSS file, the combined size of these files is 308.25 kb.	Because Foundation simply utilizes jquery 2 to optimize javascript and speed up site loading, it is fairly straightforward.

2.4 The Front End technologies will be used for the development

2.4.1 Front End Programming Language

After spending time researching about front-end programming languages, I decided to choose HTML as the main programming language for this project. Here are a few reasons why I made this decision:

- HTML is easy to learn and use: Most programmers consider HTML to be one of the computer languages with the easiest learning curve. HTML is frequently taught to developers before any other language, and it is an essential component of all development courses. HTML is straightforward to learn since it uses basic tags, and case insensitivity is not a problem.

- HTML is free: Developers do not need to purchase any software in order to begin using HTML because it is a free language. Users may access major aspects of the language without using any other plugins or applications. This is why a large number of companies choose HTML to meet their website design needs. Complete websites might be created in HTML.
- HTML offers support for multiple browsers: The most popular browsers all support HTML. So, if a website is created in HTML, there is no need to worry about browser compatibility. For the many browsers from which they may be loaded, HTML webpages can be optimized.
- Basic design: HTML is a simple language with a correct structure, specified labels, and characteristics. In this way, if I regularly practice, I will become used to the simple HTML structure and genuinely wish to write HTML code without any trouble.
- Utilization of formats: HTML permits the utilization of formats for developers. In this way, it makes the errand of the web engineers significantly simpler by diminishing the time taken to compose the plan codes.

Besides, I also use CSS design language to develop my project. Here are a few reasons I use CSS for implementations:

- Enhances user experience: Easy-to-use formatting is possible using CSS to customize user experiences. It also has a wealth of features, including animations and transitions. The consumer is not only satisfied but also delighted by this.
- Consistency: The uniformity of CSS is a noteworthy advantage. All pages utilizing that style sheet will be updated if I make a simple formatting change. To be clear, just one line of code will need to be changed for the style to be applied to all web pages. In this way, I may alter the appearance of my entire website with a few lines of code.
- CSS saves time: The CSS code only has to be written once, and it may be applied to several HTML pages. Each HTML element has a style that may be specified and depending on the situation, a user can utilize it on one or more web pages.
- Easy maintenance: Cascading sheet not only simplifies website development but also simplifies the maintenance as a change of one line of code affects the whole website and maintenance time. Users may quickly make global modifications to update the style since CSS code is simple to maintain. All web pages are immediately updated upon updates.

2.4.2 JavaScript Library / Framework

Regarding the JavaScript framework, I decided to use Angular framework to implement my project for the following reasons:

- Out-of-the-box functionality: For developers to get started, Angular comes with a ton of tools by default. Tools that may be utilized to handle routing needs are among them. It may be used by developers to retrieve the necessary data they have in an app. Development and testing are accelerated by Angular's environment's predefined setup.
- Consistency: Compared to React, Angular is a more complete and responsive web design framework. The creation of services, components, and modules is simple for developers.

- MVC architecture: Angular divides my web app into three distinct parts — Model (data), View (the UI layer), and Controller (business logic). The three units can be developed in parallel and separately tested. As a result, the code becomes easier to understand, maintain, and extend.
- Two-way data binding: The JavaScript framework uses a two-way data binding approach to synchronize the Model and the View. The View updates in tandem with the Model's data. Engineers may speed up development since no additional code is needed to maintain constant View and Model synchronization.

2.4.3 CSS Framework

To develop and design the interfaces of my web-based application, I will be using a CSS framework called Bootstrap to implement. Here are the reasons I use Bootstrap:

- Easy initiation: The main benefit of using Bootstrap in web development is that it makes coding simple. Bootstrap takes care of everything. Using Bootstrap, one can create a captivating landing page even without considerable coding skills. Additionally, both older and newer websites may easily integrate Bootstrap. I may easily include many Bootstrap platforms, frameworks, and components in my existing CSS.
- Ample documentation: The community's help is invaluable while utilizing a framework, however, some people prefer to use printed notes. Thankfully, Bootstrap fulfills this need by offering clear and effective documentation on the Bootstrap website. The documentation and simple foundation makes it simpler to work on UI design.
- Responsive Business Actions: Developers may opt to display or conceal each grid area for specific operating systems. Bootstrap comes with a 12-column grid structure that makes it simple to design for various screen sizes. The building of customized screen displays for quick mobile marketing is sped up by this responsiveness.
- Easy to Integrate: The platform's API layer may be easily integrated thanks to Bootstrap's versatility. Designs may coexist peacefully with already-existing pages, websites, apps, and software.

3 Back End technology stack

3.1 Back End Programming Language

3.1.1 C#

C# is one of the select few computer languages that has consistently placed lower than the top 5 programming languages over the past few years according to different standard indexes. However, you should be aware that Microsoft originally created this general-purpose language, particularly for the .Net framework. In addition to backend web development, C# is now widely utilized in a variety of fields, including the creation of Windows programs and video game development. The language offers you a number of beneficial characteristics, including component-oriented and structured language, quicker compilation, interoperability, scalability, and updateability. Additionally, C# offers a wide range of libraries that aid developers in creating software more quickly and effectively. C# is currently extensively used in a range of disciplines, including the production of Windows programs and video games, in addition to backend web development. Component-oriented and structured language, speedier compilation, interoperability, scalability, and updateability are just a few of the advantageous features the language provides. Additionally, C# has a huge selection of libraries that help programmers create applications more efficiently and fast (GeeksforGeeks, 2022).

```
using Windows.UI.Xaml.Controls;
using Marek.Classes;

namespace Marek.Pages
{
    public sealed partial class Hangman
    {
        internal static TextBlock TextBlock;
        internal static Grid Grid;
        internal static Image Image;
        public Hangman()
        {
            TextBlock = CopiedWord;
            Grid = GridMain;
            Image = Picture;
            Init();
        }
        public void Init()
        {
            InitializeComponent();
            Words.LoadWords();
            DisplayWord.DisplayTheWord();
            Buttons.Hangman_OnLoaded();
        }
        public void PlayAgain_OnClick(object sender, RoutedEventArgs e)
```

Figure 17: C#[15]

Features:

- Simple to learn: Because C# is a high-level language, it is more similar to other well-known programming languages like C, C++, and Java and is therefore simple for anybody to learn.
- Frequently used for creating desktop and web apps: C# is frequently used for creating desktop and web applications. It is one of the most widely used languages on business PCs. C# is the language of choice for anyone looking to develop Microsoft applications.
- Community: A larger community is preferable since it will spur the development of new software and tools. Due to the sizeable C# community, advancements are made to keep the language alive and prevent its extinction.
- Some key features of C# include cross-platform compatibility, garbage data, value collection, and object-oriented programming.
- Some widely used C# frameworks for backend development are ASP.NET Core and .NET MVC.

3.1.2 PHP

PHP, which stands for hypertext preprocessor, is a scripting language, which means that the code you write in it is read by the computer directly rather than being translated into machine-readable code by a compiler. Since PHP is an interpreted language, practically every major operating system, including Windows, Linux, macOS, Unix, etc., can run it without the need for a compiler. Additionally, PHP's deployment tools and capabilities make deploying, submitting, and validating your code much simpler. PHP is best utilized for server-side operations. PHP is beneficial in many different fields of development because HTML, a common front-end tool for developing online apps and pages, can contain PHP code. Another advantage of PHP is that it makes it easy for newcomers to see results from their back-end coding since it frequently ignores little mistakes that don't significantly effect the program's ultimate output. Even though you might have to go back over your software later to fix any errors in your code, doing this might make you feel more successful and teach you how to complete a project's back end (GeeksforGeeks, 2022).

[15] https://sitenco.com/developpement-c-sharp/

Figure 18: PHP[16]

Features:

- Cross-platform compatibility: Because it can be used on any operating system and in the Windows environment, PHP is multi-platform and renowned for its portability. XAMPP (Windows, Apache Server, MySQL, Perl, and PHP) and LAMP are the most popular (Linux, Apache, MySQL, PHP). Since PHP is platform-independent, integrating it with other databases and other technologies is fairly simple and doesn't require re-implementation. It successfully conserves a great deal of energy, time, and money.
- Active community support: PHP has a large and diversified online community of developers that may assist newcomers with developing web-based applications. These international volunteers provide fresh versions of the PHP libraries as well as other new features.
- Maintenance: Code maintenance is a crucial step in the web development process, especially when working on large projects. There are several PHP frameworks available, such as MVC (Model View Controller), which facilitates the creation and upkeep of code. The various modules' files are kept in distinct locations.
- Support for third-party applications and security: Many of PHP's built-in functions include data encryption options, making the language more safe. Users themselves can safeguard data using third-party software.
- Characteristics of object-oriented programming: PHP supports these features, which speeds up the language and adds new ones like data encapsulation and inheritance at various levels.

3.1.3 Java

Java is a well-liked language for programmers who wish to construct robust, large-scale online applications that need high levels of security to protect data. It's a flexible language that you may use to construct online, mobile, and desktop apps and tools with various digital platforms, such as computers and mobile devices. Java operates on the Java Virtual System (JVM), which standardizes the computer on which programmers

[16] https://mona.media/php-la-gi/

run code instead of enabling it to run on each programmer's particular machine, which accounts for its adaptability and reliability. Because Java has so many potential implementations, many programmers utilize it for web and software development. However, this also means that there are numerous online resources available to assist novice programmers in learning how to use Java for successful back-end projects. To put it more precisely, the Java code is first translated into machine-independent byte code, and this machine-independent byte code then executes on the JVM regardless of the underlying architecture. Java also supports multithreading, which enables the simultaneous execution of two or more threads for optimal CPU efficiency. Platform independence, adherence to OOP principles, extensive open-source libraries, automated memory allocation & garbage collection, portability, etc. are some other benefits of Java (GeeksforGeeks, 2022).

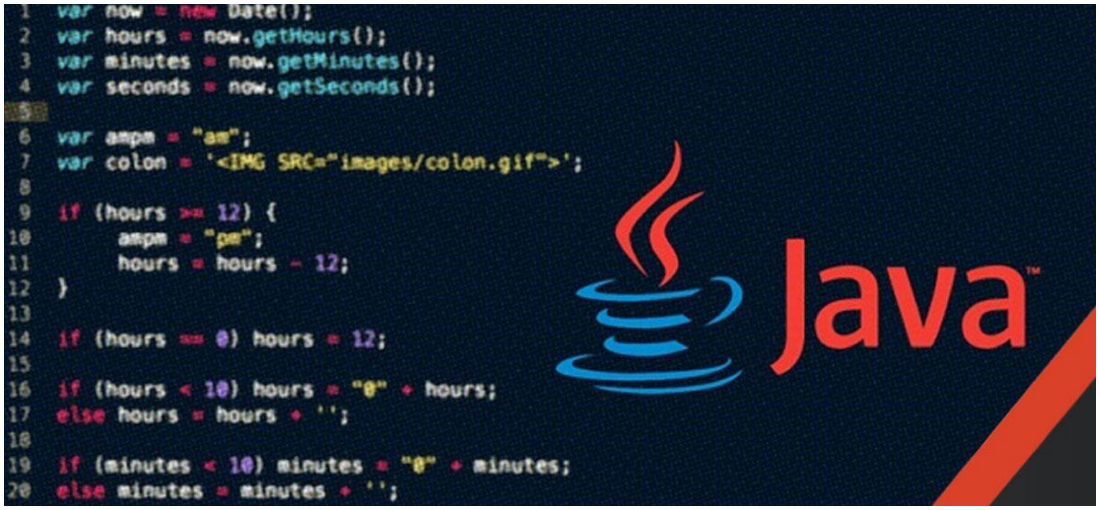

Figure 19: Java[17]

Features:

- Platform Independent: The JVM then executes the bytecode that the compiler produced after converting the source code to bytes. This means that if we develop a program on Windows, we can run it on Linux, and vice versa. This bytecode can run on any platform, including Windows, Linux, and macOS.
- Object-Oriented Programming Language: An approach to object-oriented programming is to structure the program as a collection of objects, each of which represents an instance of the class.
- Simple: Because it lacks sophisticated features like pointers, operator overloading, multiple inheritance, and explicit memory allocation, Java is one of the simpler programming languages.
- Multithreading: Java has multithreading capability. It is a Java feature that enables the simultaneous execution of two or more software components for optimal CPU efficiency.
- Robust: Java is a powerful and dependable language. As a result of its design, which prioritizes error detection as early as possible, the Java compiler is able to identify mistakes that are difficult for other programming languages to see. Garbage collection, exception handling, and memory allocation are the three primary components of Java that provide it robustness.

[17] https://topdev.vn/blog/hoc-lap-trinh-java/

3.1.4 The comparison of the back-end programming language

Table 7: C# vs PHP vs Java

Parameters	C#	PHP	Java
Speed and Performance	Faster overall	Lower performance and it does not allow threading at all.	Can be made faster using Just-In-Time compilers
Security	Many features make it more vulnerable to attacks	Is less secure, uses other frameworks for security	Very secure because it's statically typed and has code vertification
Popularity and talent pool	Though like by developers much more than Java, C# is less demanded right now	Is less popular among companies and developers	Java has a more consistent hiring and professional demand in the industry
Easiness to learn	Easy to learn	Easy to learn	Easy to learn
Features	Object - oriented and component - oriented language . Multithreading, automatic garbage collection, large library, code reuse, event management with delegates, indexers, lambda support .	Web servers & database support, understandable syntax, error handling, platform-independent, loosely typed language, and real-time access monitoring.	Multithreading, automatic garbage collection, portability and platform independence, strong security , high - level concurrences, checked and unchecked exceptions support.
When to use	C# is widely used for developing desktop applications, web applications and web services. It is used in creating applications of Microsoft at a large scale. C# is also used in game development in Unity.	The database accessibility and flexibility that PHP provides makes it one of the best backend programming languages for search engines and eCommerce platforms. PHP is also a popular choice for gaming apps, social media platforms, and remote access for businesses.	Java's security and portability make it the best language for backend for building scientific applications, enterprise solutions, as well as games. Besides that, Java is a common choice for web applications and Android mobile products.

3.2 Operating System

3.2.1 Windows OS

The Microsoft Corporation developed Microsoft Windows, a multitasking operating system that interacts with users via a graphical user interface. Traf-O-Data was the company's original name until November 1975, when it became "Micro-soft," and then on November 26, 1976, it became "Microsoft." With the introduction of Microsoft DOS version 1.0, a 16-bit command-line operating system, in August 1981, Microsoft joined the market (MS-DOS). The Windows operating system is the company's flagship product. Microsoft was founded by Bill Gates and Paul Allen (Priya Pedamkar, 2021).

Figure 20: Windows OS[18]

Features:

- Software compatibility: Windows' success is also attributable to the wide range of programs with which it is compatible. The Windows operating system is often compatible with any program you find online. Currently, a huge selection of browser applications, picture editing programs, video editing programs, and video player apps are all accessible. Another popular program that is exclusive to the Windows operating system is Microsoft Office. Other operating systems cannot be used with this program; it is only compatible with Windows operating systems.
- Hardware Compatibility: The bulk of hardware producers provide Windows compatibility with each new piece of hardware they produce. Hardware makers are unable to ignore the Windows operating system due to its immense popularity. Almost anything you purchase from a computer vendor is compatible with the Windows operating system. Additionally, these gadgets are usable as soon as they are plugged in. Windows automatically detects hardware components as long as the required drivers are set up.
- Driver Update: Device drivers, a type of software, allow operating systems and hardware to communicate with one another. Without the drivers, the hardware component won't operate correctly. The drivers may now be easily updated if you use Windows OS. The Windows operating system updates the drivers automatically when a new driver is published. If you do this, your device drivers will be up to date.

[18] https://www.educba.com/introduction-to-windows/

- High-End Gaming: Windows OS is the best choice for gamers. mostly due to its vast selection of video game titles. Windows OS is crucial to game developers in particular since it controls the market. The majority of games created for Linux and Mac also exist for Windows. The Windows OS is made for intense gaming, which makes them distinct from one another. Most games developed for previous versions of Windows will function just fine on the most recent one. Games on Xbox and PlayStation consoles compete with games on Windows PCs as a result.

3.2.2 mac OS

The macOS operating system was created by Apple. The operating system is the sole component of a computer that you can use. macOS is pre-installed on all new Apple Macintosh computers (commonly known as Macs). Thanks to macOS, you may do a number of routine tasks on your computer. For example, you may use it to play games, listen to music, edit digital photos, and browse the Internet. macOS is also used in certain companies since it gives users access to productivity software like word processors, spreadsheets, and calendars. There have been several iterations of macOS available since the middle of the 1980s. Numerous updates have since been made, with macOS Mojave (2018), High Sierra (2017), and Sierra is the most recent (2016). (2015) Yosemite and El Capitan (2014). OS X was the name of earlier versions of macOS (pronounced O-S ten). However, many people still use the terms OS X and macOS interchangeably because of the operating system's core characteristics (GFCGlobal, 2021).

Figure 21: macOS[19]

Features:

- Fewer virus attacks: In addition to having fewer active users and being the second most popular operating system, macOS also has fewer virus outbreaks. The other aspect of virus prevention is the UNIX connection, which is more secure than Windows OS.
- Good customer support: Users of Mac operating systems receive assistance more promptly than those of other operating systems. Apple employs skilled engineers who help users troubleshoot issues with its devices and software.
- Similar GUI for all the products: Mac OS shares the same graphical user interface (GUI) as other Apple products like the iPhone and iPads. Users who transition from earlier Apple devices feel at ease using macOS.

[19] https://www.hnmac.vn/cac-phien-ban-macos

- Performance and long life: Apple makes both hardware and software, creating a very strong physical link that improves performance. Excellent operating systems and hardware are used. Apple computers endure longer than other brands of PCs. Additionally, Mac computers have longer battery life.
- Default apps: Pre-installed programs like Onedrive make your computer perform slowly when you install Windows. This is not the case with macOS, which includes powerful programs that have no adverse effects on system speed. Two of the macOS programs that are pre-installed are iPhoto and iMovie.

3.2.3 Linux OS

The Linux operating system family is a group of open-source, free Unix-like systems built around the Linux Kernel. Linus Torvalds released it for use for the first time on September 17, 1991. The GNU General Public License permits the modification and distribution of the source code to anybody for commercial or noncommercial use. It is a free and open-source operating system. Originally created for home computers, Linux is today used in servers, mainframes, supercomputers, and other gadgets. Linux is used by all contemporary embedded devices, such as routers, automation controls, televisions, digital video recorders, video game consoles, wearable technology, etc. The biggest commercial success for Linux has been Android, an operating system that powers smartphones and tablets and is based on the Linux kernel. Of all general-purpose operating systems, Linux has the greatest installed base thanks to android. A Linux distribution is typically how Linux is packaged (ankita_saini, 2021).

Figure 22: Linux OS[20]

Features:

- The fact that Linux is an open-source operating system is its main advantage. This means that everyone has simple access to the source code and that you are free to add to, modify, and distribute the code with anybody without needing any additional permissions.
- In terms of security, Linux is the safest operating system available. Linux is less prone to malware than any other operating system, despite the fact that it does exist. This does not necessarily mean that Linux is 100 percent safe. Consequently, antivirus software is not required.
- Linux provides straightforward and frequent software upgrades.
- Depending on your needs or tastes, you can utilize any of the many Linux distributions that are available.

[20] https://analyticsindiamag.com/5-reasons-why-linux-os-is-a-hot-favorite-among-coders/

- Linux is freely available for usage online.
- It is backed by the whole community.
- It provides extremely robust stability. There's no need to restart it after a little interval because it seldom freezes or slows down.
- It safeguards the privacy of the user.
- The performance of the Linux system is much better than other operating systems. It successfully handles a sizable number of workers working simultaneously.
- It's appropriate for networks.
- There is a lot of flexibility with Linux. You are not needed to install the entire Linux suite; you are allowed to install only the essential parts.
- Linux is compatible with a wide variety of file formats.
- It is quick and easy to install via the internet. Any gear, even an obsolete computer system, may be used to set it up.
- The hard drive completes all tasks successfully despite its small size.

3.2.4 The comparison of the Operating Systems

Table 8: Windows OS vs macOS vs LinuxOS

Operating System	Windows OS	macOS	Linux OS
File structure	All of the user's various file types are organized by Windows using a directory structure. It has logical drives and drawers in a cabinet. Folders are likewise supplied. Papers, images, music, videos, and downloads all have common folders. All of these files can be stored in these folders, and more folders can be made. It also has files, which may be a spreadsheet or a piece of software. It may have the.txt or.jpg file extension, among others. Windows also provides a recycle bin where you may save all deleted files. It is possible to set up the recycling bin with a bigger capacity.	The file system of the MAC computer is known as MAC OS X. If you browse the hard disk of your MAC using the finder, you will see a lot of folders. The MAC root directory could encounter this when a user goes to their own MAC computer. You may inspect the file system and directory structure by going to folders like /Application, /Developer, /sbin, /tmp, etc.	Linux has a totally different file system than Windows and Mac. It was made with a special codebase. Data is kept in a tree-like organization. A single existing file tree is mounted over all of your devices.

Registry	Windows registry is the name of the main database that houses all of your computer's settings. It is responsible for maintaining a record of all user information, including passwords and device-specific information. If required, you may examine each entry, value, and even driver in the registry editor.	All application settings are kept in a series of.plist files that are housed in the various preferences folders on MAC. This.plist file includes all the properties, either in plain text or binary format. These are maintained in /Library's preferences folder.	Linux does not have a separate proprietary registry either. On a program-by-program basis, all application settings are stored under multiple users in the same file hierarchy format. Regular cleaning is also not required because these details are not kept in a centralized database.
Interchangeable interfaces	The Windows user interface was not changeable prior to Windows 8. There were some enhancements in Windows XP, but not enough. The start menu, taskbar, system tray, and Windows Explorer	A characteristic of MAC is a virtual network interface bridge. Go to system preferences to adjust the interfaces.	Linux makes it easy to switch interfaces. Changing the environment doesn't need moving all of the installations. Programs like GNOME and KDE help to satisfy these requirements. They help focus on different things.
Command terminal	Using a terminal or command prompt is the best way to execute commands. It goes by the name Windows Command Processor as well. Various batch files and commands may be run with it. Additionally, it may be used to do administrative duties and to fix any issues with Windows.	MAC provides a console as a terminal application. It has a command line, console, prompt, and terminal. You use the command line to input commands. You may use the prompt to issue commands and get some information. A terminal is a real interface that also provides a modern graphical user interface. Under Applications -> Utilities is where you may find the terminal.	Moreover, Linux includes a terminal. Applications => Utilities or Applications => System both provide access to the terminal. A shell prompt can be found in addition to this. shell in bash that is most frequently used. It establishes the behavior and appearance of the terminal upon execution.
Support framework	ASP.NET Framework, ASP.NET Core	ASP.NET Core	ASP.NET Core

Support IDE	Visual Studio Code, Visual Studio	Visual Studio Code, Visual Studio	Visual Studio Code
Price	Wide range of prices to choose from	High cost, difficult to access	Linux is developed for free for users and is based on an open source platform.

3.3 Web Server

3.3.1 NGINX

NGINX is a high-performance, open-source, free IMAP/POP3 proxy server, HTTP server, and reverse proxy. High performance, stability, a feature-rich feature set, easy configuration, and low resource usage are all attributes of NGINX. Both as a reverse proxy and a web server, it can scale effectively. Although it does not let you assign a process to a specific connection, it does establish a process pool that is readily shared by a number of connections in the network. Every time a request is made, a resource will be assigned to the process, improving resource usage and making it simple to handle several connections. Additionally, NGINX aids in establishing a secure link between your data centers and the outside network. It also works well as an HTTP load balancer that allows you to use multiple different load-sharing mechanisms (DevopsCurry, 2021).

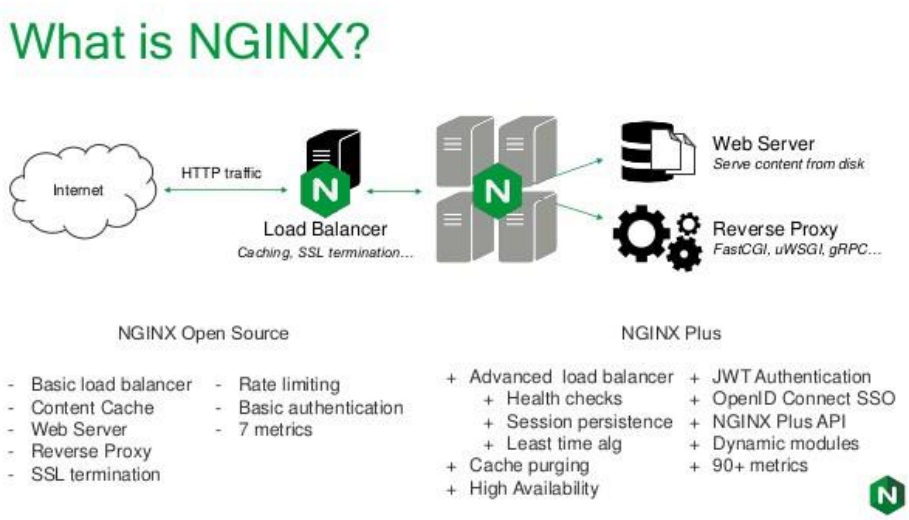

Figure 23: NGINX[21]

Features:

- Caching: For improved efficiency, the NGINX server offers a cache for both static and dynamic content. You must pay to direct each data request to the microservice that creates it. Microcaching can be used to cache the data for a brief time and lessen the stress on the back-end infrastructure.
- Consolidated logging: The standard HTTP log format is included with NGINX. Instead of keeping a separate log for each microservice's traffic and then combining them, it enables you to log all web

[21] https://medium.com/devopscurry/what-is-nginx-understanding-the-basics-of-nginx-in-2021-f8ee0f3d3d54

traffic on the NGINX front end. You can make producing and preserving access logs simpler by using NGINX.

- Scalability and fault tolerance: You may scale your back-end infrastructure so that adding or deleting any microservice will not have an impact on the user's experience using NGINX's load balancing and health checks capabilities.
- Zero downtime: The webserver's seamless operation is guaranteed by NGINX. Even the system software can be seamlessly updated or upgraded without affecting the connection, preventing any application downtime.
- Mitigate DoS attacks: NGINX is well-known for handling tons of incoming requests or HTTP traffic, ensuring application safety during high traffic, common cache response, and deliver request smoothly.

3.3.2 Apache

American software developer Robert McCool is the author of the Apache open-source Web server. As a Web server, Apache is in charge of approving directory (HTTP) requests from users of the Internet and delivering their requested data in the form of files and Web pages. The capabilities of Apache are intended to be compatible with a large portion of the Web's software and code. Apache is frequently used at home by programmers creating Web applications to preview and test their code. The Apache software also offers a safe and secure file-sharing function that enables users to store files in the root directory and share them with other users. The distinctive license under which software from the Apache Software Foundation is released accounts in part for the Apache server's influence on the open-source software ecosystem. Because it has so many modules that increase the capabilities of the web server, Apache, a web server created in the C programming language, is extremely adaptable. In addition to many more, they include mod file cache for caching, mod ftp to enable FTP for file uploads and downloads, and mod ssl to handle SSL/TLS encryption protocols (The Editors of Encyclopaedia Britannica, 2022).

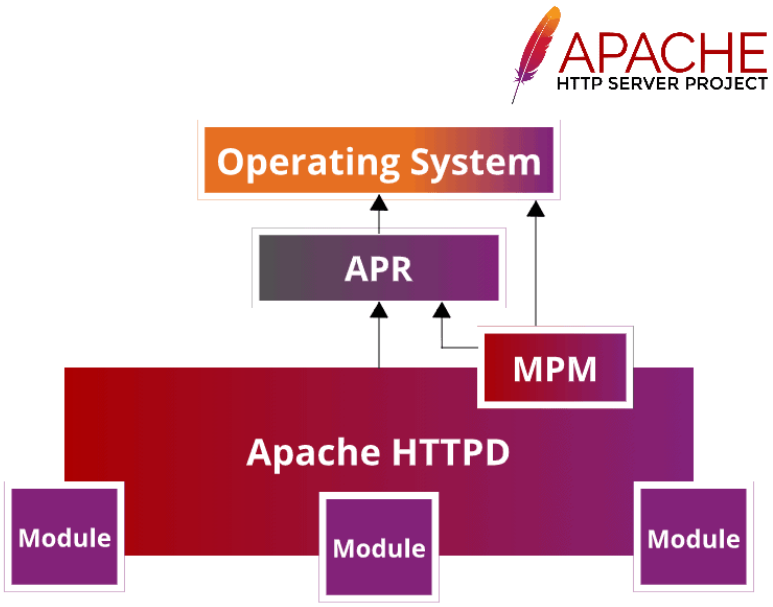

Figure 24: Apache[22]

[22] https://phoenixnap.com/kb/nginx-vs-apache

Features:

- Unix Threading: Apache httpd is now capable of operating in a hybrid multiprocess, multithreaded manner on Unix systems with POSIX threads capability. This enhances scalability for many setups, but not all of them.
- New Build System: The build system has been completely redesigned to use autoconf and libtool as its foundation. As a result, the configuration structure of Apache httpd is more comparable to those of other packages.
- Multiprotocol Support: Apache HTTP Server now has some of the infrastructure in place to support serving multiple protocols. mod_echo has been written as an example.
- IPv6 Support: Apache httpd receives IPv6 listening sockets by default on platforms where the underlying Apache Portable Runtime library supports IPv6. Additionally, IPv6 numeric address strings are supported via the directives Listen, NameVirtualHost, and VirtualHost.
- Multilanguage Error Responses: SSI documents are currently used to send error response messages to the browser in a number of different languages. The administrator can alter them to create a unique appearance and feel.

3.3.3 The comparison of the web server

Table 9: NGINX vs Apache

	Apache	NGINX
Overview	Open-source cross-platform web server	Open-source cross-platform web server used as a reverse proxy, load balancer, mail proxy and for HTTP caching
Architecture	Process Deiven Approach Creates a new thread for each request Hence multi-threaded	Event-Driven approach Handles multiple request within one thread Lightweght structure, and much faster architecture
Performance (Both static and dynamic content)	Serves static content using the file-base method, which is an overhead for the interpreter Processes dynamic content within the server	Static content can be served in a straight-forward manner and the interpreter will only be contacted when needed It does not process dynamic content and passes to an external processor for execution
OS Support	Supports all Unix-like system including Linux and BSD. It fully supports MS-Windows	Supports almost all Unix-like OS Support Windows partially

Distributed / Centralized Configuration	Allows additional configuration on a per-directory basis via .htaccess files.	Does not allow additional configuration
Security	Greate security	Better security with the smaller codebase
Flexibility	Supports customization of web server through dynamic modules	Not flexible enough to support dynamic modules an loading
Feature Modules	60 official dynamically loadble modules that can be turned On/Off	3rd Party core moddules (not dynamically loadable)

3.4 Database

3.4.1 My SQL

MySQL is a quick and easy RDBMS that is used by both small and large enterprises. MySQL is developed, promoted, and supported by MySQL AB, a Swedish company. There are several reasons why MySQL is becoming more and more well-liked, including (Tutorialspoint, 2018):

- A collection of tables with linked data is referred to as a database.
- A table is a type of data matrix. A database table resembles an easy-to-use spreadsheet.
- Only data of the same kind may be included in a single column (data element), such as the postcode column.
- A row is a group of related data, often referred to as a tuple, entry, or record, such as the details from a single subscription.
- To speed up a system, redundancy is the practice of storing data twice.
- A primary key is a unique item. A key value cannot occur twice in the same table. With a key, just one row may be found.
- A foreign key connects two tables by acting as a connecting line.
- A compound key, also known as a composite key, is made up of several columns since just one column provides enough uniqueness.
- Index The index in a database is comparable to the back index in a book.
- There can never be referential integrity without a foreign key value pointing to an already-existing row.

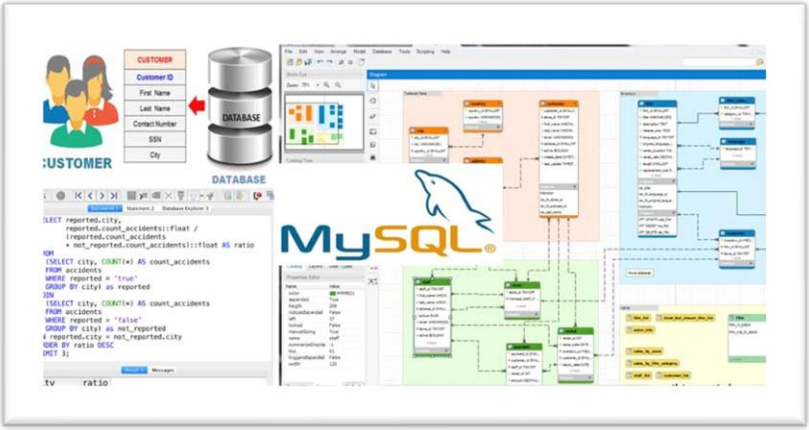

Figure 25: MySQL Database[23]

Features:

- Data Security: WordPress, Drupal, Joomla, Facebook, and Twitter are just a few of the well-known web apps that use MySQL, the most dependable and secure database management system. Any firm, especially one that operates an eCommerce store with regular money transfers, may gain a lot from the data security and transaction processing capabilities offered by MySQL's most recent version.
- On-Demand Scalability: MySQL has unmatched scalability to make managing deeply embedded programs with a smaller footprint easier, even in massive warehouses that stack terabytes of data. MySQL's flexibility on-demand is a key feature. eCommerce businesses with specific database server requirements can entirely modify their systems using this open-source solution.
- High Performance: Thanks to its distinctive storage-engine topology, system administrators may configure the MySQL database server for flawless performance. MySQL is designed to handle even the most demanding applications while assuring maximum speed, full-text indexes, and specialized memory caches for increased performance, whether it's a high-speed transactional processing system or an e-commerce website that receives a million queries per day.
- Round-the-Clock Uptime: With MySQL, a variety of high-availability alternatives are available, including the promise of continuous uptime (24/7), such as dedicated cluster servers and master/slave replication setups.
- Comprehensive Transactional Support: MySQL is presently the most dependable transactional database engine available. Due to features like full atomic, consistent, isolated, persistent transaction support, multi-version transaction support, and unrestricted row-level locking, it is the best alternative for comprehensive data integrity. It ensures quick deadlock identification through the server's enforcement of referential integrity.
- Complete Workflow Control: With an average download and installation time of less than 30 minutes, MySQL enables usefulness from day one. Regardless of your platform being Linux, Microsoft, Macintosh, or UNIX, MySQL is a comprehensive solution with self-management tools that automate everything from space expansion and setup to data creation and database administration.

[23] https://wikimaytinh.com/co-so-du-lieu-la-gi-mysql-la-gi-database.html

- Reduced Total Cost of Ownership: Businesses who move from their current database software to MySQL see considerable cost savings on new projects. The time spent troubleshooting performance and downtime issues can be reduced because to the dependability and ease of management.
- The Flexibility of Open Source: All doubts and concerns about choosing an open-source solution may be removed with MySQL's 24/7 support and corporate indemnity. MySQL's dependable software and safe processing deliver efficient transactions for high-volume applications. While making maintenance, troubleshooting, and upgrading quick and easy, it enhances the end-user experience.

3.4.2 MongoDB

MongoDB, the most popular NoSQL database, is a cost-free document-oriented database. NoSQL is short for non-relational. It implies that MongoDB does not rely on the relational database structure that looks like a table and instead provides a whole other mechanism for data store and retrieval. BSON is the brand name of this storage format (similar to JSON format).

SQL databases are used to store tabular data. The predetermined data model in which this data is stored is not particularly adaptable for the contemporary, quickly growing real-world applications. More than ever, applications nowadays are social, interactive, and networked. Applications are storing and retrieving data at accelerating rates. Relational database management systems (RDBMS) are not the greatest choice for processing enormous amounts of data due to their architecture, which prevents them from scaling horizontally. If the database just utilizes one server, it will run into scaling issues. NoSQL databases are more scalable and provide greater performance. MongoDB is a NoSQL database that expands by adding more and more servers while increasing productivity thanks to its flexible document architecture (GeeksforGeeks, 2021).

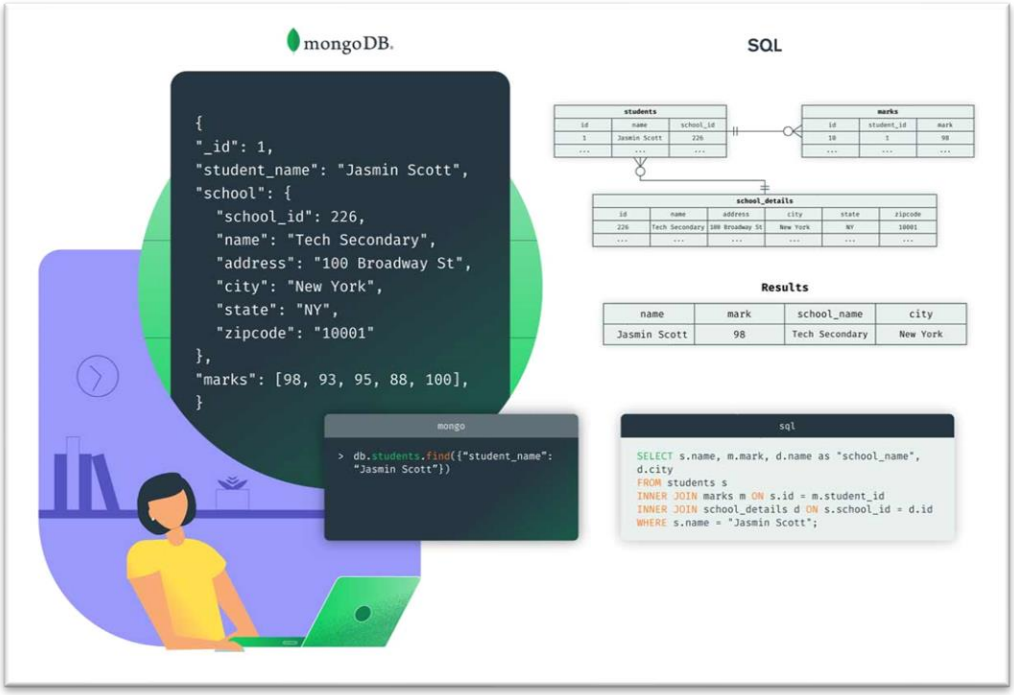

Figure 26: MongoDB[24]

[24] https://prodima.vn/mongodb-la-gi/

Features:

- Document-Oriented: As opposed to fragmenting the principal topic into several relational structures like RDBMS does, MongoDB stores it in the fewest number of documents possible. For instance, it contains all of the data about a computer in an one document titled Computer rather than storing it in many relational structures like CPU, RAM, Hard disk, etc.
- Indexing: Without indexing, a database would need to spend time and resources scanning every document in a collection to find the ones that match the search parameters. Since efficient searching depends on indexing, MongoDB uses it to swiftly examine huge amounts of data.
- Scalability: Sharding enables horizontal scaling in MongoDB (partitioning data across various servers). Data is divided into pieces and then evenly distributed among shards that are deployed over a number of physical servers using the shard key. New machines can be added to an active database.
- Replication and High Availability: Data availability is improved via MongoDB, which makes numerous copies of the data accessible on various servers. By providing redundancy, it protects the database against hardware malfunctions. If one server goes down, the data may be simply accessed from other active servers that also have the data saved on them.
- Aggregation: Operations that aggregate data records process the records and deliver the calculated results. It is comparable to the SQL GROUPBY clause. Sum, average, min, max, and other aggregation expressions are a few.

3.4.3 The comparison of the database

Table 10: MySQL vs MongDB

Database	MySQL	MongoDB
Use Case	Apps from the past or those that demand multi-row transactions (i.e. accounting systems)	Internet of things, smartphone apps, and real-time analytics.
Data Structure	Clear template for structured data	It is not necessary to define a schema.
Risk	SQL injection attack danger	Less risk of attack due to design
Analysis	An excellent choice if you need a traditional relational database and have ordered data	An excellent choice if your organized and/or unstructured data has the potential to grow significantly.
Type	RDBMS	Document-Oriented
Main points	- Table - Row - Column	- Collection - Document - Field
Schemas	Strict	Dynamic
Key features	- Full text searching and indexing - Integrated replication support - Triggers - SubSELECTs - Query caching - SSL support - Unicode support - Different storage engines with various performance	- High write loads - Unstable schema - Your DB is set to grow big - Data is location based - HA (high availability) in unstable environment is required - No database administrators (DBAs)

3.5 Hosting

3.5.1 Azure

Microsoft's public cloud computing platform is called Microsoft Azure, formerly known as Windows Azure. Numerous cloud services are offered by it, such as computation, analytics, storage, and networking. These services are available for users to select from while creating and scaling new apps or using the public cloud to operate already existing applications. The Azure platform intends to assist organizations in managing difficulties and achieving their objectives. It is interoperable with open source technology and provides solutions that assist many industries, including e-commerce, banking, and a number of Fortune 500 organizations. This gives people the freedom to use the tools and technology of their choice. Azure also provides four main types of cloud computing: serverless, platform as a service, software as a service, and infrastructure as a service. Azure is billed by Microsoft on a pay-as-you-go basis, which means customers are only billed for the exact resources they have actually utilized (Stephen J. Bigelow, 2020).

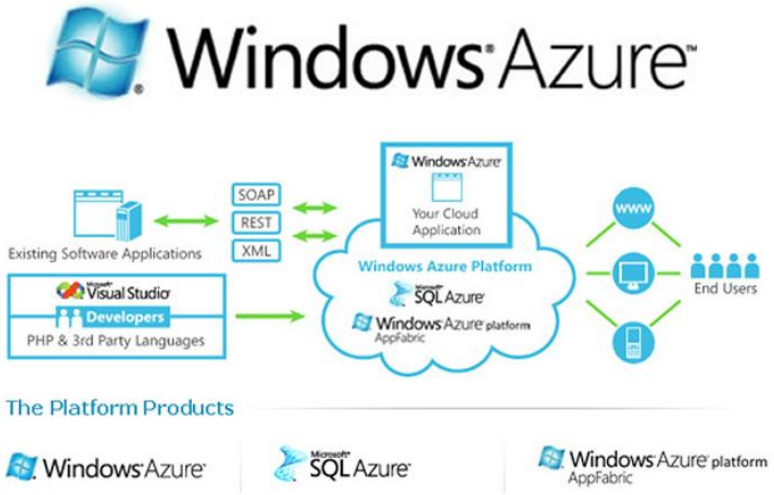

Figure 27: Azure[25]

Features:

- Enterprise-grade security: Azure's high level of security is one of the main factors influencing why businesses want to use it. Microsoft dedicates more than a billion dollars annually to cybersecurity research and development. Additionally, it has a staff of more than 3,500 international cybersecurity specialists that take care of the clients' security demands. Users are also informed by Azure when upgrades are necessary. Users may safeguard their data by using multi-factor authentication and strong passwords. The proactive Azure Security Center keeps an eye on user activity around-the-clock, 365 days a year, and takes the required action if it detects a risk to user data.
- Reliable disaster recovery: No vendor of cloud computing services can provide error-free operation and 100 percent uptime. The reputation of the platform, however, depends on how soon the issue is remedied. One such platform with a cutting-edge disaster recovery mechanism is Azure. It provides a variety of solutions for disaster recovery that support users in trying circumstances.

[25] https://blog.vsoftconsulting.com/blog/what-the-heck-is-windows-azure/

- Speedy development and delivery: Users can quickly design, test, and deploy new apps because to Microsoft Azure's emphasis on speed. Additionally, it allows users to analyze a lot of data rapidly and restore data from backups swiftly. Azure makes use of robust templates and automated solutions to improve the user experience.
- Focus on cost optimization: Azure is a cloud computing platform that can be used by both small and large businesses. This is due to the subscription plan's predictable cost and transparency. It continually configures autoscaling and turns off unnecessary resources to provide reasonable pricing. Users must pay for what they use after purchasing the beginning bundle, which starts at $13/month.

3.5.2 AWS

Amazon's cloud computing platform, AWS (Amazon Web Services), is extensive and constantly expanding. It combines infrastructure as a service (IaaS), platform as a service (PaaS), and packaged software as a service (SaaS) products. An enterprise may benefit from AWS services by receiving resources like computing power, database storage, and content delivery services. Amazon.com developed internal infrastructure to manage its online retail activities, and AWS was introduced in 2006. One of the first businesses to provide a pay-as-you-go cloud computing model that grows to meet users' demands for computation, storage, or throughput was Amazon Web Services (AWS). For businesses and software developers, AWS provides a wide range of tools and solutions that may be utilized in data centers in as many as 190 nations. Groups such as government agencies, education institutions, nonprofits and private organizations can use AWS services. AWS can be broken into three main products: EC2, Amazon's virtual machine service, Glacier, a low-cost cloud storage service, and S3, Amazon's storage system (Alexander S. Gillis, 2020).

Figure 28: AWS[26]

[26] https://www.netsolutions.com/insights/what-is-amazon-cloud-its-advantages-and-why-should-you-consider-it/

Features:

- Cost Savings: With AWS, companies pay for what they use. There's no upfront cost to build a storage system and no need to estimate usage. AWS customers use what they need and their costs are scaled automatically and accordingly.
- Scalable and Adaptable:Start-ups and small companies can see the clear advantages of utilizing Amazon for their computing needs because AWS's cost is adjusted based on the use of the clients. AWS really makes it easy to establish a business from scratch since it offers all the tools required for doing so. Amazon offers affordable migration services for already-established businesses, allowing for a smooth transition of your current infrastructure to AWS. AWS offers tools to support growth as a business expands. Customers won't ever have to worry about whether or not they need to reassess their computer utilization because the business model supports flexible usage.
- Security and Reliability: A corporation maintaining its own website or storage is definitely far less safe than using Amazon Web Services. Numerous data centers operated by AWS are presently spread out all over the world and are properly maintained. A disaster affecting one area won't result in a permanent loss of data due to the diversity of the data centers.

3.5.3 The comparison of the hosting

Table 11: Azure vs AWS

	AWS	Azure
Computing power	AWS leverages EC2 tailored to customer needs.	Azure users are given the option of creating a VM from a virtual hard disc (VHD).
Cloud storage offerings	AWS has services like AWS S3, EBS, and Glacier.	Azure Storage Services offers Blob Storage, Disk Storage, and Standard Archive.
Privacy and security enforcement methods	AWS performs an excellent job of selecting secure alternatives by default, ensuring enhanced privacy.	Azure is protected by a dedicated Cloud Defender service, an AI-powered tool.
Ease of use and documentation model	AWS provides a feature-rich and user-friendly dashboard, along with extensive documentation.	Azure keeps the account information in one place, although its documentation system is less intuitive and search-friendly.
Licensing and license mobility	AWS licenses are more configurable and feature-rich.	Azure provides more software as a service (SaaS) features and is easier to set up for Windows administrators.
Networking	Users can build private networks using AWS' VPC.	Azure relies on a VPN

Machine learning (ML) modeling	AWS' SageMaker gives total freedom and flexibility in creating ML models.	Azure ML Studio is primarily focused on providing a codeless experience through drag-and-drop elements.
Logging and monitoring	SageMaker logs model metrics and historical data via CloudWatch.	Azure ML Studio utilizes MLFlow for data recording and monitoring.
Database capabilities	Amazon RDS is a database service that supports six standard database engines.	Azure's SQL Server Database and Cortana Intelligence Suite supports Hadoop, Spark, Storm, and HBase.
Open-source presence	AWS is ideal for open-source developers as it is compatible with Linux and provides integrations for open-source applications.	Microsoft Azure is still in the process of embracing the open-source community.
Processes for deploying applications	AWS offers Elastic Beanstalk, Batch, Lambda, container services, etc., but it lacks a few features in terms of app hosting	Azure offers a variety of app deployment options, including cloud services, container services, functions, batches, and app services, among others.
Containerization and orchestration support	Containerized apps in AWS run using Elastic Beanstalk, which supports Docker files through a command-line interface.	In Azure, the same functionality is performed by App Service, but the process is slightly more complex as one must run the container inside of a web app.
Pricing model	AWS is billed on an hourly basis.	Azure is billed as per a per-minute model.

3.6 Framework

3.6.1 .NET Framework

The.NET software framework was conceived and developed by Microsoft. In 2002, the.Net framework's 1.0 first release became accessible. It is a virtual computer that may be used to assemble and run programs written in a variety of languages, including C#, VB.Net, etc. It is used in the development of online services, form-based applications, and web applications. VB.Net and C# are the two most popular programming languages supported by the.Net platform. It allows for the creation of applications for Windows, mobile devices, the web, etc. It supports industry standards and provides a variety of functions. The.NET Framework supports more than 60 programming languages, including 11 that Microsoft built and produced. The other non-Microsoft languages are supported by the.NET Framework even though they weren't developed by Microsoft and aren't maintained by them (Anshul_Aggarwal, 2022).

Figure 29: .NET Core[27]

There are three significant phases of the development of .NET technology.

- OLE Technology: OLE is one of the technologies included in the component definition from Microsoft (Object Linking and Embedding). Essentially, its purpose is to link together various program components.
- COM Technology: Different software components may connect with one another thanks to the COM (Common Object Model) technology used by the Microsoft Windows operating system family. Developers frequently use COM to build reusable software components, join components to build applications, and make advantage of Windows services. COM objects may be created in a number of different programming languages.
- .NET Technology: .NET technology, a collection or group of technologies, is utilized in the creation of Windows and online applications. the technical aspect. A fundamental description of Net is that it refers to Microsoft's newly unveiled Internet Strategy, which was unveiled in February 2002. It was first called NGWS (Next Generation Web Services). One of the strongest, most well-liked, and most useful types of Internet technology now accessible is said to be this one.

Main Components of .NET Framework:

- Common Language Runtime is the essential and virtual machine element of the.NET Framework (CLR). The code is executed by the run-time environment of the.NET Framework, which also makes the development process easier by providing a variety of services like remoting, thread management, type safety, memory management, resilience, etc. It is effectively in charge of managing how.NET applications are executed, independent of the.NET programming language that is being used. Since runtime-targeting code is known as managed code and non-targeting code is known as unmanaged code, it also helps with code management.
- A framework class library is a set of reusable, object-oriented class libraries, methods, and other parts that may be connected with the CLR (FCL). The Assemblies is another name for them. C/C++ header files and Java packages are pretty similar to it. Installing the.NET framework primarily involves installing CLR and FCL into the system. An overview of the.NET Framework is given below.

[27] https://vitinhquan7.info/sua-loi-yeu-cau-cai-dat-net-framework-3-5/

3.6.2 ASP.NET

The ASP.NET web application framework was developed by Microsoft. ASP.NET, an open source version of the original ASP, is a component of the.NET Framework (Active Server Pages). Version 1.0 of the.NET Framework was released in January 2002. The topic of what technology we were using to develop internet applications and services before the year 2002 thus becomes relevant. The answer is to use classic ASP. So, before.NET, there was Classic ASP and ASP.NET. Programmers may run the code for ASP.NET using any.NET language (C#, VB, etc.), as it is built on the CLR (Common Language Runtime). It is designed primarily to work with HTTP and for web developers to construct dynamic web pages, web apps, websites, and online services because it provides a great combination of HTML, CSS, and JavaScript (Anshul_Aggarwal, 2019).

The.NET Framework is used to create a large number of programs and services, including Console, Web, and Windows ones. However, ASP.NET is only used to create online apps and services. As a result, ASP.NET is referred to as a component of the.NET Framework.

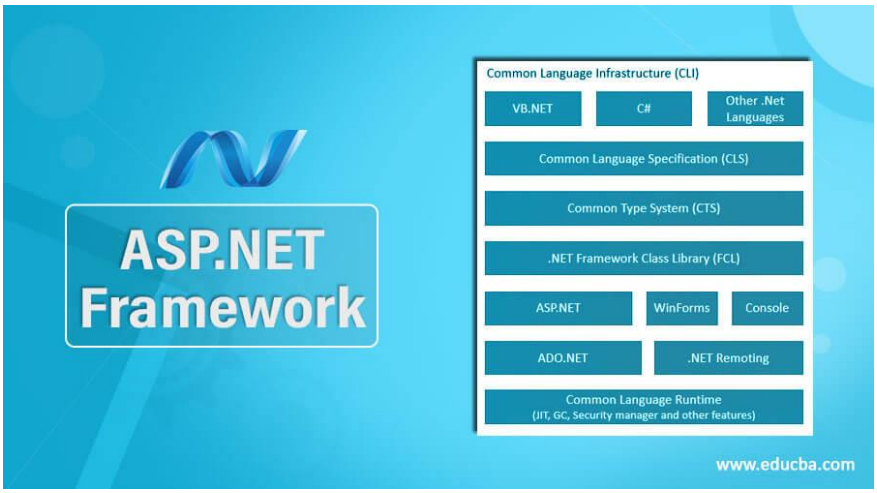

Figure 30: ASP.NET framework[28]

According to Anshul Aggarwal (2019), the following are advantages of ASP.NET Framework:

- Extending.NET Framework: Since it adds various libraries and development tools to the.NET Framework to support the building of web applications, ASP.NET is a subset of the.NET Framework. Some of the things it offers to the.NET Framework include libraries for well-liked web designs like MVC, Editor Extensions, the foundation framework to handle online requests, and web-page template syntax like Razor.
- Performance: It outperforms other web frameworks on the market in terms of speed.
- Backend Code: C# can be used to develop backend code for data access and any other functionality with the aid of ASP.NET.
- Dynamic Pages: The syntax for generating dynamic web pages with C# and HTML is provided by Razor in ASP.NET. JavaScript includes the React and Angular SPA frameworks and may be combined with ASP.NET (Single Page Application.)

[28] https://www.educba.com/asp-dot-net-framework/

- Support for many OS: The creation and execution of ASP.NET applications are supported on Windows, Linux, Docker, and MacOS. The Visual Studio provides the tools required to develop.NET programs for a variety of OS.

3.6.3 The comparison of the framework

Table 12: The comparison of the .NET and ASP.NET

.NET Framework	ASP.NET Framework
A software development framework called .NET is designed to create applications for Windows, the Web, and servers.	The .NET Framework's primary tool, ASP.NET, is designed to make it easier to create dynamic websites.
The .NET framework allows for the development of both server-side and client-side applications.	Because ASP.NET is integrated with the .NET framework, you can only develop server-side web applications using it.
Primarily employed in the creation of Windows-based business applications.	Using .NET languages, it is used to create dynamic web pages and websites.
Any language with a CIL (Common Intermediate Language) compiler can be used to program it.	You can program it in any language that is .NET compliant.

3.7 The Back End technologies will be used for the development

3.7.1 Back End Programming Language

Regarding backend programming languages, I have reviewed and evaluated existing languages to make a choice that I will use the C# language to implement my project. The following are the reasons why I chose C# as the main backend programing language for my project:

- Faster development time: Developers can code more quickly using C# thanks to a number of advantages. A language that is statically typed and simple to learn, a syntax that resembles an enlarged version of Java, and a sizable library chock full of high-level capabilities are a few of these qualities.
- High scalability: Because C# is a static coding language, all of its programs are dependable and simple to modify. Engineers may swiftly modify any C# software to increase its functionality and serve more users by building on top of it.
- Object-oriented: The language that best makes use of object-oriented programming may be C# because of the manner it has embraced it. In truth, C# is exceptionally efficient and adaptable due to its object-oriented nature, which makes programming simpler and less resource-intensive.
- Gentle learning curve: C# is a high-level language that is simple to learn and comprehend. And that's without taking into account the numerous built-in functions, all of which are incredibly user-friendly. Additionally, any engineer who is already familiar with C++ or Java will be at ease when learning C# because these two programming languages have a lot in common in terms of functionality and approach.
- Big community: Because C# is one of the most commonly used languages in the world, there are a lot of C# developers available to assist me with it. Not just that. As a Microsoft product, C# enjoys the backing of the industry leader in technology, which translates to professional assistance, additional resources, and regular upgrades.

3.7.2 Operating system

I will choose Windows operating system as my main operating system for the development of my project for the following reasons:

- Backing for all equipment: As windows OS is utilized by 95% of clients so the majority of the equipment merchants make drivers for windows.
- Convenience: Every version of Microsoft Windows has a consistent feature that makes it easy for users to switch from one version to the next. Customers of Windows 7 don't have any problem upgrading to Windows 10 because a huge number of its features are the same as those of Windows 7. Windows has an even simpler user interface than UNIX and Mac.
- Programming support: The Windows stage is best suited for programmers and game developers. Designers seek to create utilities, games, and programming for the Windows OS because of the platform's large user base. Since Linux users cannot create Windows apps, it is wiser to use Windows to do so.
- Fitting and play highlight: The majority of equipment may be identified by attachment and play incorporate naturally. Although I don't need to physically set up the equipment, it is ready to use once it is linked, such as a camera, console, mouse, mobile phone, and so on.
- Work area and contact screen: Windows 10 is made for both touch screen gadgets and PCs. The UI of Windows 10 is made so that it turns out better for a windows gadget.
- User Friendly: When compared to other operating systems, the Windows OS's Graphical User Interface (GUI) makes its operations significantly simpler. I don't need to be an expert in computers to use this operating system. This can be used by even those who have only rudimentary computer skills. Because of this, Windows users do not think of switching to another operating system.

3.7.3 Web Server

I will choose Apache as the main web server to deploy for my project for the following reasons:

- Price: Because Apache Web Server is free to download and use, it is considered open source software. I can adjust its source code to meet my own requirements. Without giving up any functionality, this provides Apache a sizable edge over nearly all of its rivals.
- Easy to Setup: New websites are often created by people who have no prior expertise with web design. As a result, the industry has shifted toward simpler, code-light web development platforms like WordPress. For novices, Apache is the ideal web server software. For most CMS, it works right out of the box and is simple to install.
- Huge Userbase: Because there is strength in numbers, as a community grows, so do the available resources. The greatest support is provided for Apache because it is the most extensively used web server software. Beginners can quickly locate the solutions to their inquiries. Many people take this for granted, but if I utilize another web server, I'll have a lot tougher time locating the answers.
- Virtual hosting support: The capability of hosting several websites on my web server, each with their own virtual directory or folder and using the same domain name. This capability is useful because it allows web developers to host several websites with various functions on a single web server, which may save them time and money.

- Domain Name System (DNS) support: I can map IP addresses to domain names without having to make a separate file because Apache supports DNS lookups. This is incredibly beneficial since it makes the process of building a website much simpler and quicker as I only need to tell Apache what the URL of my site should be rather than having it hardcoded into a different file someplace else online.

3.7.4 Database

After researching and evaluating existing databases, I have decided to use MongoDB for my project for the following reasons:

- MongoDB is schema-less: To categorize and arrange the data in relational databases, we must define tables, schemas, and relations. The development of tables and other hard, pre-defined structure is not necessary with MongoDB, on the other hand, as it is a schema-less database. MongoDB keeps all of the records in a single collection because it is a document database.
- MongoDB has no joins: RDBMS makes it difficult to link two or more tables since joins are needed. We can easily link other documents with MongoDB since it lacks complicated joins and has an easy-to-change document structure.
- No primary key set-up: Setting up a primary key is required when using an RDBMS. There is no need to explicitly construct a primary key in MongoDB. Every document in the NoSQL database comes with a _id field that is automatically constructed. As a main key, this newly generated field is used. As the main key in MongoDB, the reserved _id field needs to contain only one unique value. It is crucial to remember that MongoDB automatically inserts "MongoDB Id Object" into the _id column if it is not set up.
- MongoDB Uses CAP Theorem: RDBMS emphasizes the ACID (Atomicity, Consistency, Isolation, and Durability) qualities. For database deployments, MongoDB, on the other hand, employs the CAP theorem, which stands for Consistency, Availability, and Partition Tolerance.
- Scalability and Transactionality: One of MongoDB's greatest benefits is its capacity to scale; with its scale-out design, I can create apps that can easily handle traffic surges. This implies that the job is divided across multiple cheaper and smaller machines. Massive read and write operations can be enabled because of MongoDB's various advancements. In MongoDB, objects may be nested within one another during data modeling. In this case, updating may be accomplished with a single transaction as opposed to numerous transactions like in traditional relational databases. Additionally, MongoDB supports database transactions, which allow a batch of database changes to be merged and either accepted or refused.

3.7.5 Hosting

Regarding hosting, I will use AWS as the main hosting for my project for some of the following reasons:

- Easy to use: AWS is made to enable suppliers, ISVs, and application providers to swiftly and securely host my apps, whether they are SaaS-based or not. To access AWS's application hosting platform, utilize the AWS Management Console or well-documented web services APIs.
- Flexible: I may choose the web application platform, programming language, operating system, database, and other services I require with AWS. I get a virtual environment through AWS that I may fill with the programs and services my application needs. As a result, current applications may be more easily migrated while still having alternatives for developing new solutions.

- Cost-Effective: There are no long-term contracts or upfront payments; I simply pay for the computing power, storage, and other resources that I really utilize. Visit the AWS Economics Center for further details on comparing the expenses of other hosting options with those of AWS.
- Reliable: I can benefit from a scalable, dependable, and secure global computing infrastructure with AWS, which serves as the virtual foundation for Amazon.com's multi-billion dollar online company and has been refined for more than ten years.
- Scalable and high-performance: My application may scale up or down depending on demand using AWS technologies like Auto Scaling and Elastic Load Balancing. I get immediate access to computation and storage resources thanks to Amazon's extensive infrastructure.
- Secure: AWS utilizes an end-to-end approach to secure and harden our infrastructure, including physical, operational, and software measures. For more information, see the AWS Security Center.

3.7.6 Framework

I will use the ASP.NET framework to use for my project implementation for the following reasons:

- It allows for separation of concern: The MVC design, which is used by ASP.NET, enables for the application's input, processing, and output to be separated. Model-View-Controller is a three-tiered architecture with related components that may perform particular software application development tasks.
- Reduces coding time: When creating large applications, the framework technology is very useful for speeding up the development process. I have no opportunity of producing poor code because there are several distinct forms of code reviews. I might increase code quality by doing code reviews.
- Delivers power and flexibility: Because the framework language is built on the CLR, any web application developers may benefit from the flexibility and power of the whole platform. Additionally, it is language agnostic, allowing I to select the language for my application or even split it up into many languages.
- Simplicity: Every activity, from the most straightforward to the trickiest most convoluted, may be completed with ease. The common language runtime streamlines the development process with tools like automated reference counting and trash collection. The framework enables me to create user interfaces with distinct presentation and application logic.
- Customizability and Extensibility: Developers greatly benefit from the framework's factored architecture. With the aid of my own specially created components, I may quickly replace or enhance the subcomponent of the ASP.NET runtime. Putting things into practice has been much simpler.
- Security: Security is a good feature of the framework language. I can develop secure applications through built in Windows authentication and per-application configuration features.
- Manageability: The text-based hierarchical configuration structure of the framework contributes to its good management characteristic. I may simply utilize the local administration tools to implement the new settings because these configurations are included as simple texts. With no need to restart the server, deploy them individually, or change running compiled code, chores become more simpler as a result.

4 Tools for source control management

4.1 GitHub

A web-based interface called GitHub makes use of Git, an open source version control system that enables many users to simultaneously make distinct modifications to web pages. Carpenter draws attention to the fact that GitHub promotes real-time team collaboration while teams are creating and updating the content for their websites. It is best for distributed version control across multiple teams (Software Testing Help, 2022).

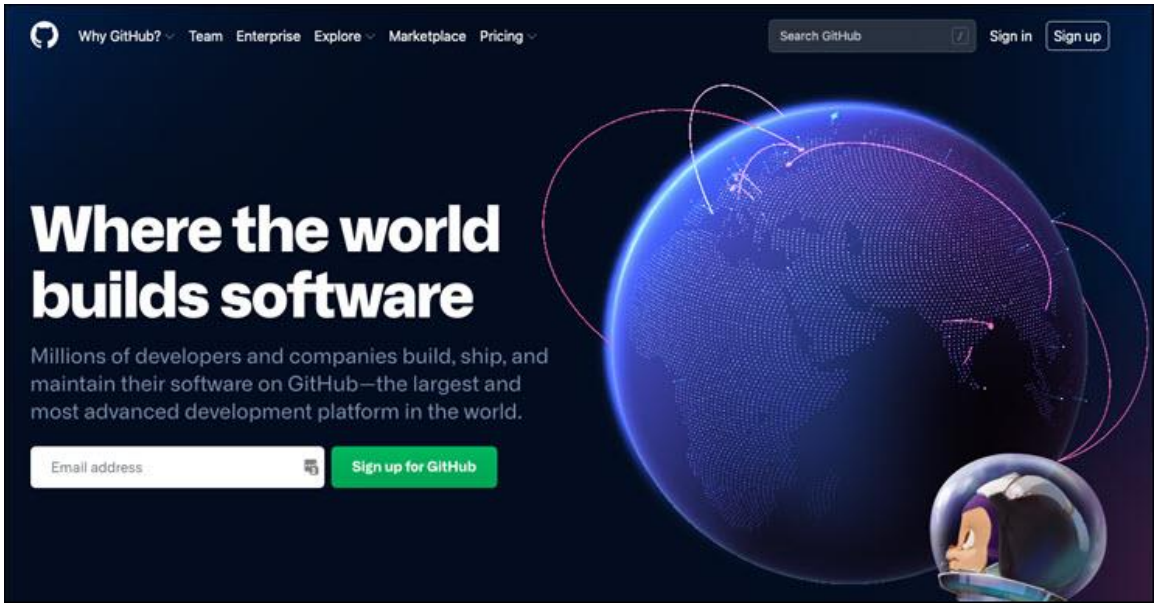

Figure 31: GitHub Tool[29]

Features:

- Allows large development teams to collaborate, evaluate, and manage oftware/application code.
- Integrates with bug tracking software.
- Access management and permissions are already supported.

Pros:

- Supports mobile devices, Macs, and Windows.
- Supports deployment on the cloud.
- Able to precisely handle various projects and permissions.
- Great assistance and documentation.

Cons:

- Due to the fact that Github is a cloud-based platform, security for high-value intellectual property projects or scripts may be an issue.

[29] https://www.softwaretestinghelp.com/best-source-code-management-tools/#2_Git

- Large pull requests may be time-consuming and labor-intensive to examine.

Pricing: Limited support is offered during the free trial period for people and organizations. The Team edition costs $4 per user per month, and Enterprise costs $21 per user per month.

4.2 GitLab

The web-based Git repository known as GitLab includes a free open and private repository, issue-following tools, and wikis. It is a thorough DevOps platform that gives specialists the ability to handle every project task, from planning and maintaining the source code to monitoring and security. Teams may work together to produce better software. GitLab helps teams work more effectively and accelerate product lifecycles, which benefits customers. It is not necessary for program users to manage tool authorizations on an individual basis. If permissions are simply provided once, then everyone in the organization has access to every component (Karin Kelley, 2022).

Users have the option of choosing GitLab's commercial edition if they want access to extra capabilities. For instance, the monthly user price for the Premium edition is $19.

Figure 32: GitLab tool[30]

Features:

- It enables project management, CI/CD integration, and Git workflows.
- There are more than 30 million individuals and 100,000 companies using it.
- For the entirety of the DevOps life cycle, a single application.

Pros:

[30] https://thinhweb.com/tong-quan-ve-git-va-gitlab-va-tai-sao-nen-dung-gitlab/

- Uses the built-in Git version control system.
- Simple to learn and utilize.
- Good assistance (depending on the pricing tiers).

Cons: Larger teams and organizations may find it less financially viable due to high prices.

Pricing:

- Free to all people.
- The prices for the Premium and Ultimate editions are $19 and $99 per user/month, respectively.

4.3 The comparison of the tools for source control management

Table 13: GitHub vs GitLab

Parameters	GitLab	GitHub
Developed by	Dmitriy Zaporozhets and Valery Sizov created GitLab.	Chris Wanstrath, Tom Preston-Werner, P. J. Hyett, and Scott Chacon created GitHub.
Open-sourced	GitLab's community edition is available for free.	GitHub is not an open source project.
Public Repository	Users are able to create public repositories.	Users are granted unlimited free repository access.
Private Repository	Additionally, GitLab offers a free private repository.	Users can create a free private repository on GitHub, but only with a limit of three contributors.
Navigation	GitLab offers the capability of repository navigation.	Users can navigate usability on GitHub.
Project Analysis	Users can view project development charts through GitLab.	The commit history can be checked, however GitHub does not yet offer this feature.
Advantages	- GitLab's community edition is open source and freely downloadable. - It is a very secure cloud-native application.	- It aids in the organization of the project document that we produce. - It is utilized to present the work to an audience.
Disadvantages	GitLab is available but contains a lot of issues, which degrades the user experience. For newcomers, managing code reviews is challenging.	A small private repository exists. Only Git version control is supported. A small private repository exists. Only Git version control is supported.
Company	GitLab Inc.	Microsoft co.

Security	More secure than GitHub	It is less safe since GitHub lacks a security dashboard and license compliance.

4.4 Tools for source control management to be used for project development

I've chosen to utilize GitHub as our project's source control management technology for the following reasons:

- Sincerely, GitHub is used to coordinate the development of practically every open-source project. Using GitHub is cost-free if my product is open source and features a wiki and issue tracker that make it simple to publish more thorough documentation and get feedback on my project. If you want to contribute, just clone a project, make my changes, and then use the GitHub web interface to send a pull request.
- I use GitHub to make it easier to get high-quality documentation. The help pages and lessons for git cover almost any topic I can think of.
- GitHub is the most useful tool I have for this. Nowadays, the vast majority of companies look at GitHub profiles when hiring new employees for their project. If my profile is made public, even if I did not attend a prominent university or college, my chances of landing a job will rise.
- I can use a common text editor to produce stylish documents with Markdown. The use of Markdown across the board on GitHub, even in user comments and the issue tracker, has revolutionized writing. With so many various computer languages to learn for submitting projects, it is unquestionably a huge advantage to have submissions made in a format without the need to learn yet another system.
- Although it has previously been mentioned, GitHub is a repository. This suggests that it enables folks to view my work. The fact that GitHub has grown to be one of the biggest coding communities has also increased the visibility of my project.
- When numerous individuals are contributing to a project, it may be difficult to keep track of who edited what, when, and where those files are stored. GitHub addresses this problem by keeping track of each change that has been submitted to the repository. To avoid losing older versions of my code with each iteration, I might keep a version history of it, similar to how you would with Google Drive or Microsoft Word.
- Known systems like Amazon and Google Cloud can communicate with GitHub, which can also track my comments using tools like Code Climate and highlight the syntax of more than 200 different programming languages.

5 Software Development Models

5.1 Waterfall Model

Software development is divided into predetermined chunks using the sequential waterfall model. Stages cannot overlap; the end of one must come before the beginning of the next. Each step of the SDLC is designed to complete a specific goal. It was introduced in 1970 by Winston Royce (Matthew Martin, 2022).

The Waterfall Model is a linear method of developing applications that uses predetermined phases that don't cross over. Because the procedures are sequential, it is impossible for developers to return to earlier stages if the model is left unmodified (thus the name "waterfall": Once water falls, it cannot rise again) (Eric Conrad, 2011).

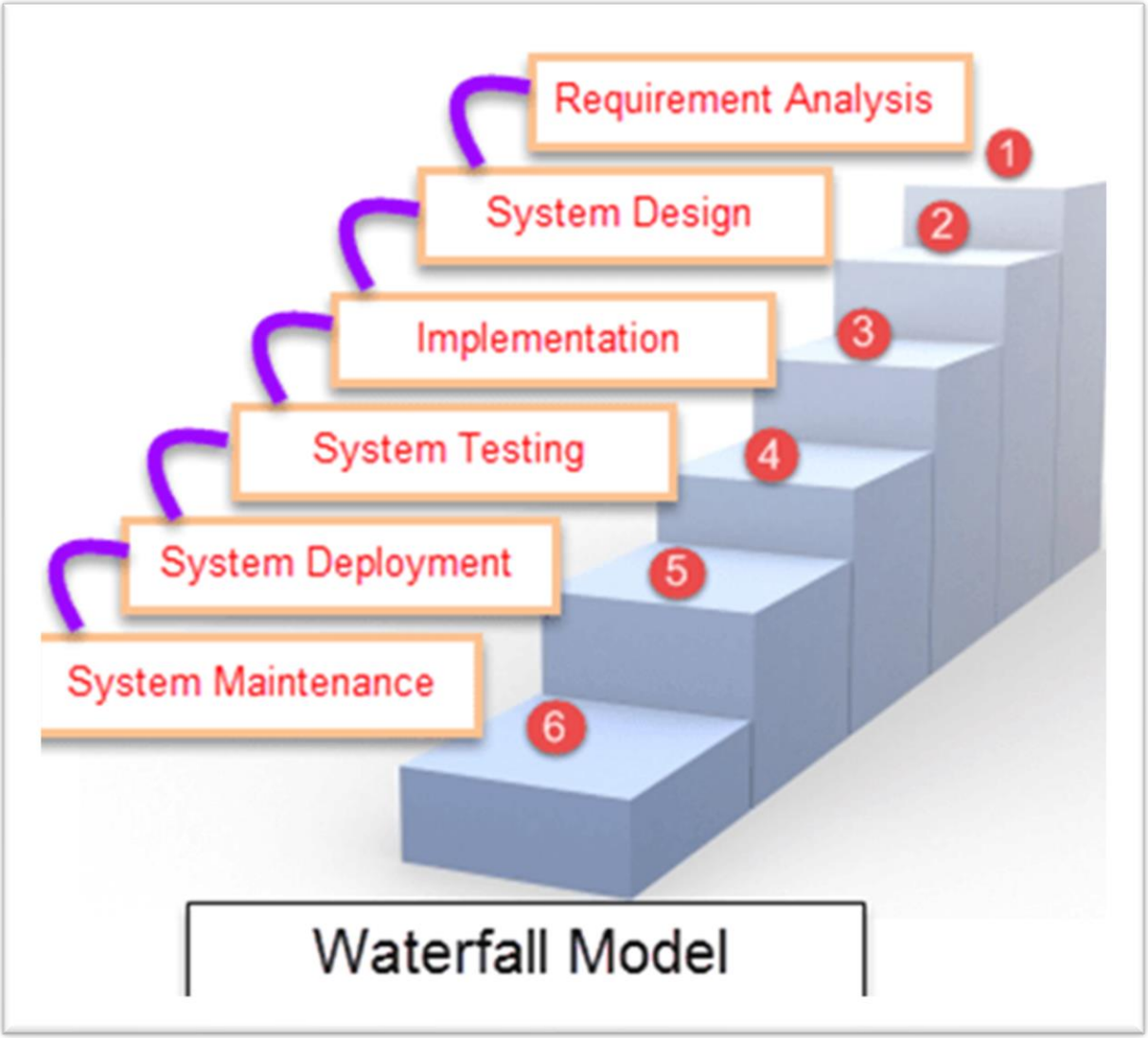

Figure 33: Explain Waterfall model in SDLC[31]

According to the website Hestabit (2021), the following are the Waterfall Model phases:

- Requirement Analysis:
  - In this stage, the client provides information on the specific requirements for the software system that will be created.
  - Recognizing and accurately capturing the precise requirements of the consumers.

[31] https://www.guru99.com/what-is-sdlc-or-waterfall-model.html

- Recording all the performance, connectivity, and connection needs with the client and developer.
- Analyzing the requirements step-by-step and conducting a brainstorming session.
- Creating Software requirement Specification (SRS), a comprehensive document outlining in plain words what the system will do.
- In this stage, the "what" and "how" of the system are discussed.

- System Design:
  - Plan the programming language, such as Java, PHP, or.net
  - or the database, such as Oracle, MySQL, etc.
  - or other high-level technical project details.
  - Putting the criteria gathered in the SRS in a format that enables for additional programming in a programming language.
  - A thorough design that describes the whole program architecture.
  - Design documentation, also referred to as a Software Design Document (SDD).
- Implementation:
  - The software is only coded at the built stage, which follows the design stage.
  - Putting the idea into practice using little programs known as "units," as per the SDD.
  - Through unit testing, the functionality of the codes is tested.
  - Including the codes in the following stage.
- System testing:
  - During this stage, you test the program to ensure that it was created in accordance with the client's specifications.
  - Incorporating the proven unit of code into a full system
  - Testing both functional and non-functional aspects to make sure the system satisfies the requirements
  - Evaluating the final product's quality via extensive system testing.
  - If any irregularities are discovered, report the problems.
  - The use of technologies like ALM, traceability matrices, etc. to monitor the testing process.
- System deployment:
  - Introduce the application to the appropriate environment.
  - Putting the finished, tested product into use with the appropriate customer or on the market.
  - After software deployment, running a sanity check on the environment to see if it is working properly.
- System Maintenance:
  - Once your system is operational, you might need to update the code in response to consumer requests.
  - Confirming that the software operates well in the relevant environment.

- After the software has been deployed and installed, giving the customers maintenance and operational support.
- Noting and resolving any problems that the clients may have had.
- Adding the newest features to the system.

In waterfall development, analysts and users move gradually from one phase to the next. The key deliverables for each phase as the project moves forward are frequently long (sometimes hundreds of pages) and submitted for approval to the project sponsor and approval committee. When the product of a phase is approved, the subsequent phase begins. As the project moves from phase to phase, it advances in a waterfall-like pattern. It is challenging to go back through the steps (from design to analysis, for example), however it is doable. (Consider swimming upstream in a waterfall like a salmon.) (Alan Dennis, 2012).

Utilizing a waterfall development technique has the advantage of allowing for the identification of requirements prior to the start of programming and limiting requirement modifications as the project advances. The key negatives are that programming can't begin until the design is finalized, that delivering the system that was suggested during the analysis phase takes a while, and that testing is almost never taken into account during the implementation phase. Since deliverables are typically a poor way of communication, it's possible that important needs may be overlooked in the volume of paperwork. If the project team misses a vital need, expensive post-implementation programming may be needed. Users may forget what the system was designed to achieve given the length of time between its initial conception and its actual execution. A system that fit the environmental criteria during the study phase could need significant modification to do so when it is put into operation in the fast-paced business world of today. Starting at the beginning and going through each phase one at a time while making the necessary revisions is required to finish this rewrite (Alan Dennis, 2012).

Advantages of Waterfall Model:

- One benefit of waterfall development is the capacity for departmentalization and control. By establishing deadlines for each step and adhering to a plan, a product may progress through the stages of the development process model one at a time. Design, implementation, testing, installation, troubleshooting, operation, and maintenance come after conceptualization in the development process. Developmental phases are carried out in a specified order.
- Straightforward and simple to use
- Because of the model's rigidity, it is simple to manage. Specific deliverables and a review process are included in each phase.
- Each phase is processed and finished separately.
- Works well for smaller projects with clearly defined requirements.
- There are distinct stages.
- Easily understood benchmarks
- Simple to organize tasks
- Results and processes have been well-documented.

5.2 Spiral Model

The Spiral model was initially introduced by Barry Boehm in 1986. This model adapts as we learn new things. In a word, a spiral model is one where we repair our faults or blunders after each step is tested. It is the key element of the SDLC (Software Development Lifecycle Model). The Spiral Model is divided into phases. These phases make up only a small fraction of the entire software development process. At the end of each part, each phase is checked. These phases can occur in any order. It varies from one job to the next. The Spiral Model's phases are organized according to design objectives, and the process is completed with a client evaluation. As a result, this strategy creates a product in a series of incremental steps. The number of task regions where each phase is activated is large (Priya Pedamkar, 2019).

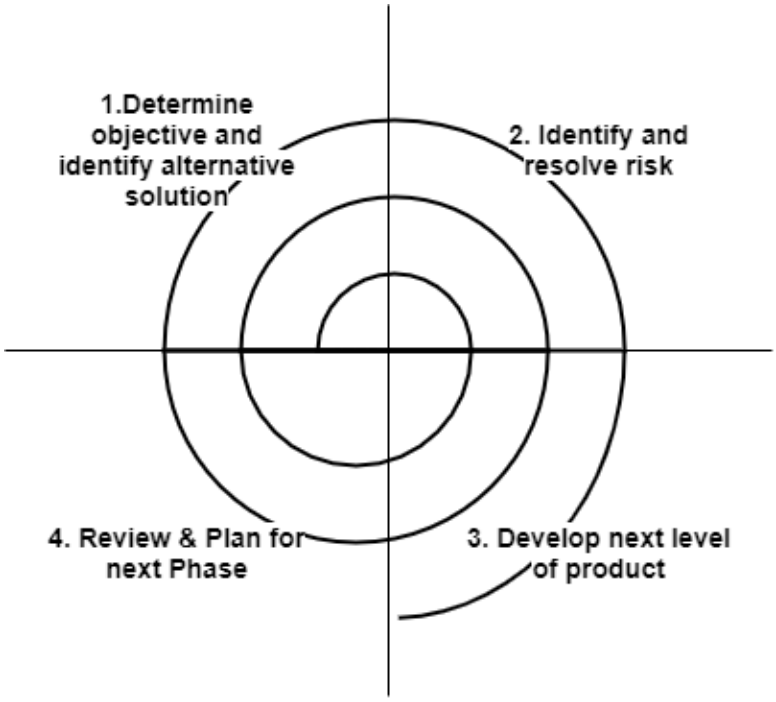

Figure 34: Spiral Model[32]

A risk is an unexpected result that might jeopardize a software project's successful conclusion. This risk can be reduced by building an access subsystem prototype and testing with the precise access rate. At every level of the software development process, this paradigm makes it possible to create a prototype, immediately aiding in risk management (Easytechnotes, 2021).

According to Priya Pedamkar, this model is divided into four phases or quadrants which are as:

- Determine Objective: Determine Objective: The first step in this phase is to gather all the criteria. As our product grows, this step verifies any system needs. This collection of criteria includes two separate categories. Business requirement specifications (BRS and SRS) are the same thing (System requirement specifications).
- Identify and resolve the risk: It's crucial to identify risks early. Only address the problem at this point. If a danger materializes, a different course of action is proposed and implemented. Additionally, the prototype was developed at this time.

[32] https://easytechnotes.com/spiral-model/

- Development and test: The primary stage of this model is this. Product testing and development are finished at this point. Testing has also been carried out after software creation.
- Planning the next iteration: The subsequent iteration begins here.

Advantages of Spiral Model:

- The model cycles. It functions in a spiral pattern, similar to a linear waterfall model.
- This model is based on risk.
- They make use of prototypes.
- This paradigm is far more adaptable than the waterfall model.
- It needs documentation in order to be maintained.
- Large projects are managed using strategic methods.
- Every stage is in our hands.
- Depending on what the customer wants, we may adjust and make modifications.
- The time it takes to create software could be shorter than expected.
- Decreased risk factor
- The end user is given a glimpse of the system.
- It is a continuous process.

Disadvantages of Spiral Model:

- It is inappropriate for the modest and low-risk product since it might be costly for a smaller enterprise.
- As many unnecessary intermediate steps as possible are present.
- The spiral model makes management somewhat difficult, thus it is a difficult process.
- Both a review process and parallel delivery are not required under the spiral approach.
- Testing was only done by developers due to the traditional nature of the business.

5.3 Incremental Model

One of the most popular models of the software development process is the incremental model, which separates the software requirement into several distinct modules for the software development life cycle. Following the division of the modules, incremental development will be carried out in phases, covering the essential analysis, designing, implementation, testing, and maintenance. Because each step is produced individually, each iteration stage of the software development life cycle will go incrementally through the requirements, design, coding, and testing modules (Priya Pedamkar, 2019).

Requirements are divided into multiple distinct software development cycle modules when adopting the incremental paradigm for software development. The processes of requirements, design, implementation, and testing are all applied to every module in this architecture. Every time a new version of the module is released, its functionality grows. The procedure is carried out until the whole system is established (JavaTpoint, 2020).

Object-oriented methodologies emphasize gradual and iterative development that is constantly tested throughout the project. With each iteration, the system becomes increasingly responsive to the demands of the users (Alan Dennis, 2012).

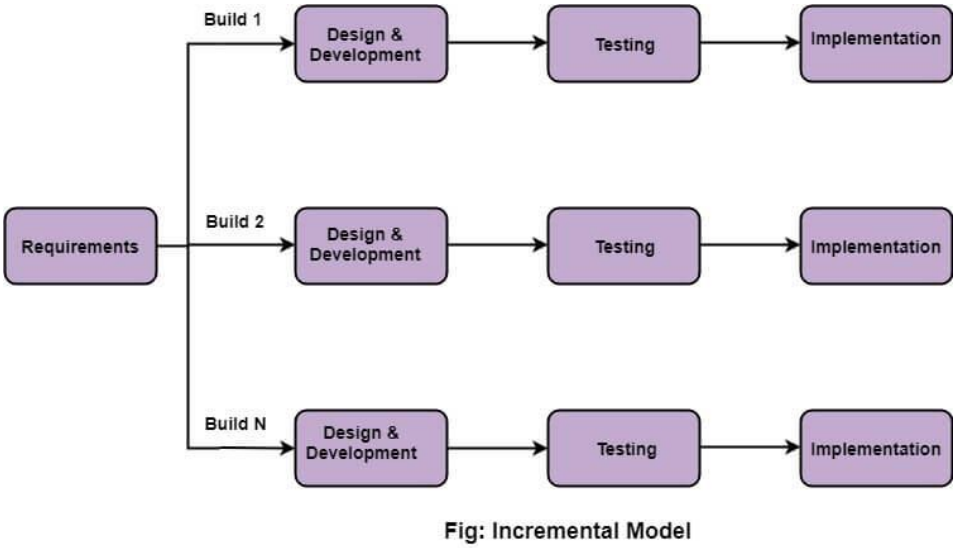

Figure 35: Incremental Model[33]

According to the website JavaPoint (2020), the various phases of incremental model are as follows:

- Requirement analysis: At the beginning of the incremental model, the needs are determined by the product analysis expertise. The functional requirements for the system are also understood by the requirement analysis team. For the incremental model of software development, this stage is crucial.
- Design & Development: At this stage of the incremental SDLC model, the system functionality design and the development approach are successfully finished. When software gains additional functionality, the incremental approach takes style and development phases into account.
- Testing: The incremental model's testing step assesses the efficiency of each current function as well as recently introduced capability. Throughout the testing process, a variety of methodologies are used to test the behavior of each task.
- Implementation: The system installation phase enables the coding stage. Final coding is included in the planning and development phase, while functionality testing is included in the testing phase. After this stage is over, the number of working products is increased and enhanced all the way to the final system product.

Advantages of Incremental model:

- Errors are simple to identify.
- Easier to test and troubleshoot
- More adaptable
- Risk is easily managed because it was dealt with during its iteration.

[33] https://www.javatpoint.com/software-engineering-incremental-model

- Important functionality is delivered to the client early.

Disadvantage of Incremental Model:

- Planning is necessary
- Cost overall is expensive.
- Module interfaces must be clearly defined.

According to Priya Pedamkar (2019), the main advantage of the incremental approach is that it divides software development into more manageable components, each of which is created utilizing the SDLC's Analysis, Design, Code, and Test stages. This strategy will guarantee that the program achieves all of its objectives, regardless of how small the target may be. So, by employing this strategy, we are able to fully accomplish the software's objective. We are also able to guarantee that the final product is free of errors and that each stage is compatible with previous and later stages of development since we carefully test at the end of each step. Now let's look at some of the benefits of the incremental model and why it is so well-liked.

- As a result of the software development being broken up into several stages, there will be a ton of little software sub-projects.
- To accomplish the software's final purpose, the unfinished systems that will be built will be merged.
- Every action or demand will be assigned a priority level, and the highest demand for the software will be taken care of first.
- Additionally, once a prerequisite for an increase has been satisfied, that specific increase will be postponed while the prerequisites for the next increment are taken into consideration.
- There won't be any long-term objectives in the plan; it will only be outlined for the present increment at a certain period. So, all of the focus will be on the need that is being worked on.
- Each incremental version is produced after the processes of analysis, design, coding, and testing, and it is frequently produced using an iterative waterfall methodology. These modifications might potentially be produced using different models.

5.4 The comparison of Software Development Models

Table 14: Waterfall Model vs Spiral Model vs Incremental Model

Properties of Model	Waterfall Model	Incremental Model	Spiral Model
Planning in early stage	Yes	Yes	Yes
Returning to an earlier phase	No	Yes	Yes
Handle Large-Project	Not Appropriate	Not Appropriate	Appropriate
Detailed Documentation	Necessary	Yes, but not much	Yes
Cost	Low	Low	Expensive

Requirement Specifications	Beginning	Beginning	Beginning
Flexibility to change	Difficult	Easy	Easy
User Involvement	Only at beginning	Intermediate	High
Maintenance	Least	Promotes Maintainability	Typical
Duration	Long	Very long	Long
Risk Involvement	High	Low	Medium to high risk
Framework Type	Linear	Linear + Iterative	Linear + Iterative
Testing	After completion of coding phase	After every iteration	At the end of the engineering phase
Overlapping Phases	No	Yes (As parallel development is there)	No
Maintenance	Least Maintainable	Maintainable	Yes
Re-usability	Least possible	To some extent	To some extent
Time-Frame	Very Long	Long	Long
Working software availability	At the end of the life-cycle	At the end of every iteration	At the end of every iteration
Objective	High Assurance	Rapid Development	High Assurance
Team size	Large Team	Not Large Team	Large Team
Customer control over administrator	Very Low	Yes	Yes

Properties of Model	- When quality comes before cost or deadline. - When specifications are precise, known, and well-defined. - The existing product has to be updated. - Transferring an existing product to a new platform	- In low- to moderate-risk projects. - The need to provide essential features as soon as feasible - On projects with lengthy development schedules. - While working on a project using new technology, it is best to let the user grow acclimated to the system gradually rather than switching overnight. - Because developing the complete system at once is quite harmful.	- For tasks with a moderate to high degree of risk. - When evaluating costs and risks is important. - When significant changes are expected. - When users are unsure about their specific needs.
Planning in early stage	Poor	Good	Good
Returning to an earlier phase	Poor	Good	Poor
Handle Large-Project	Good	Good	Good
Detailed Documentation	Good	Good	Good
Cost	Poor	Excellent	Excellent
Requirement Specifications	Poor	Excellent	Good

5.5 Software Development Models to be used for project development

Here are a few reasons why we chose to employ the Waterfall model for software development given its advantages and appropriateness:

- It is uncomplicated, understandable, and easy to use.
- It is easy to manage.
- If the demands are clearly stated, it works well for smaller, less expensive initiatives.
- stages that are clearly identifiable and defined.
- It is easy to organize your tasks.
- Processes and results have been thoroughly recorded.

- Requirements: A single document that is used to verify each stage combines the requirement, constraints, and functional and non-functional needs of the project.
- Design: The design was provided in line with the requirements of the project and was based on those requirements. The designer can be contacted to check the exact project requirements.
- Implementation: Although the original concepts were modified over time to suit the demands of the product, the criteria are evident in the documentation. Before implementation is complete, a new design must be created and approved.
- Verification: The designed and produced solutions are then evaluated against the basic criteria to confirm that the project is successfully meeting them. The problems must be re-verified if not.
- Maintenance: Either problems may worsen or the product will require an upgrade. There will be adjustments.
- Each phase in the waterfall model must be finished before going on to the next.
- The project is small, but the requirement is clearly defined and articulated.

The criteria of the FPT Book are well known, clear, and established. The system description is stable, and the technical stack is explicit. The Waterfall methodology also includes very little client involvement throughout product development. The final output can only be seen by the end consumers. As a result, the waterfall design method was used to build the FPT Book.

6 Illustrate all your findings on how to use these by drawing the overview

The following is an overview of my system, including the technologies and techniques that I have chosen to develop for the project combined and worked together. The diagram is divided into two parts, the left side corresponds to the front-end technologies that I have chosen to develop the visual aspects of the website – the part that users see and interact with, and the right side is the technologies that I have chosen to deploy the Back-end development comprises the site's structure, system, data, and logic.

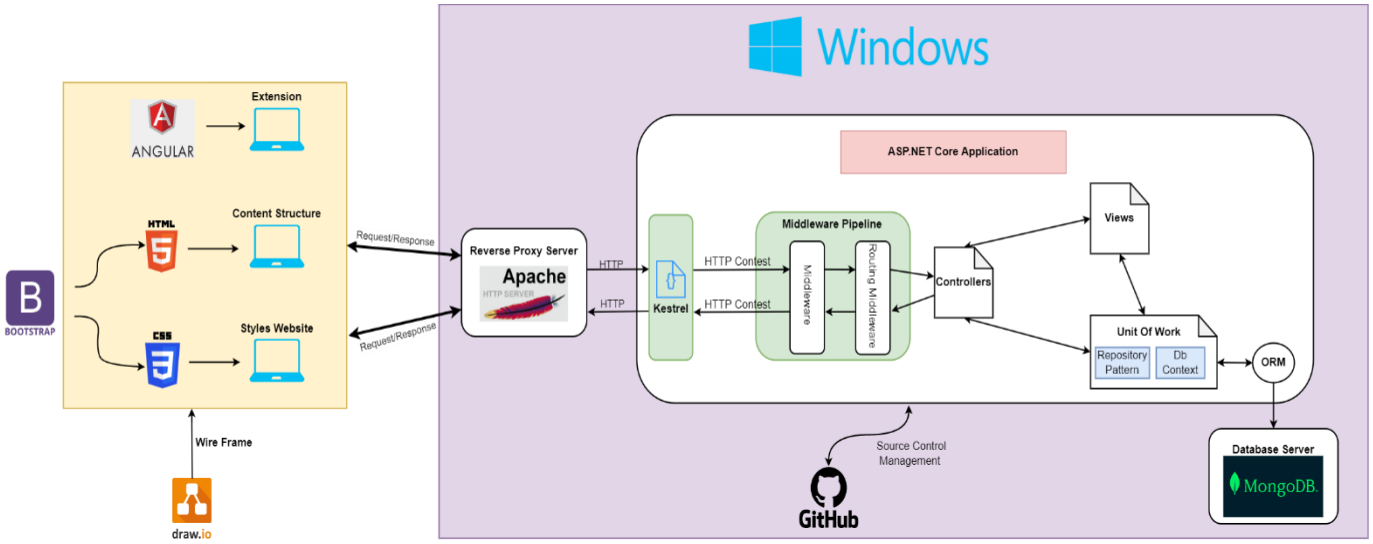

Figure 36: Illustrate overview

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